

Perceive, Externalize, Synthesize: Why Two-Pass Decomposition Beats End-to-End Multimodal Review

Bojie Li
Pine AI

Noah Shi
University of Washington

Abstract

The default way to put a multimodal model on a long-form review task is to hand it the raw source (the audio, the page images) and let it reason end-to-end. We give counter-intuitive evidence that this is the wrong default. **Decomposing the task into two same-weights passes is systematically more faithful and more complete:** one pass transcribes the source to text, and the other writes the review from that transcript alone. This holds even though the second pass sees strictly *less* than the end-to-end pass. We then ask *why*, with a seven-experiment mechanistic study, and find the cause is not what one might assume. It is **not** a perception failure. The model’s own Pass-1 transcript covers 99.9% of probes, and 361/362 (99.7%) of the probes the end-to-end pass drops are present in that transcript. The model perceives the content, then loses it. It is also **not** that text is an intrinsically better reasoning substrate. On an open VLM, identical propositional content is read equally well from text and from an *image* of that text, for a single fact and for sixteen packed facts alike (a layer-wise logit-lens shows no readout-depth gap). The cause is a **generation-side bottleneck**. A single pass cannot simultaneously perceive, reason, and write a long faithful review, so it satisfices and drops roughly a third of what it saw. Decomposition helps because it gives perception a dedicated externalization pass and synthesis a dedicated, full-budget pass over complete text. Collapsing the two into one call (transcribe-then-review in a single response) crashes coverage or fails to emit a review at all, because one output budget cannot hold both. We measure the effect on a 21-cell suite on Gemini 3.1 Pro spanning 11 audio sources (26–85 min) and 10 papers (9–176 pp), with cross-vendor extensions to Claude Opus 4.7 and MIMO v2-omni, two-judge replication (GPT-5.4 and Claude on all 21 cells), $n=5$ multi-seed verification on four headline cells, and probe-circularity and reference-bias ablations. Decomposition reduces hallucinations on 18/21 cells (mean -3.86 , sign-test $p = 0.0015$) and improves coverage on 16/21 (mean $+0.118$, $p = 0.027$). The gain scales inversely with the end-to-end baseline ($r = -0.45$, bootstrap 95% CI $[-0.73, -0.10]$): the more the monolithic pass satisfices, the more decomposition recovers. The boundary is equally informative. Decomposition stops paying off in two failure modes. The first is a *compression bottleneck* (Mode A), when the synthesis pass cannot fit the transcript into its output budget. The second is a *training-prior leak* (Mode B), where, with the raw source gone, the synthesis pass confabulates from prior knowledge (replicated on Claude across four citation-heavy papers). A quote-grounded **iterative-refinement variant** (C_{liter}) re-grounds the synthesis pass and flips the Mode-B Gemini counter-cell to a Pareto win ($n=5$ mean 7.8h/0.54cov $\rightarrow$ 3.6h/0.66cov), though on longer cells it trades coverage for faithfulness or fails outright. Perceive, externalize, then synthesize. The win is decomposition, not the modality.

1 Introduction

Multimodal foundation models can read audio and document images directly. The intuitive way to deploy them on a long-form review task is therefore end-to-end: feed the model the raw source and ask for the review in one call. Keeping the original modality in context looks strictly safer: no transcription step to lose information, and the model sees everything. We argue, and measure, that this intuition is backwards for long-form review.

The alternative is a same-weights two-pass decomposition: the model first transcribes the source to text (Pass-1, *perceive and externalize*), then writes the review from that transcript alone (Pass-2, *synthesize*, with the same weights and the raw modality removed from context). This wins. The obvious explanation, that text is simply a better substrate for the model’s reasoning, turns out to be wrong. We run a seven-experiment mechanistic study (§4) that isolates the cause:

- **Not perception.** The model’s own Pass-1 transcript covers 99.9% of probes. Of the 362 probes the end-to-end pass drops, 361 (99.7%) are present in that transcript. The model perceives the content, then loses it downstream.
- **Not substrate.** On an open VLM, a layer-wise logit-lens finds identical propositional content is read equally well from text and from an *image* of that text, whether one fact or sixteen packed facts, with the same answer-token log-prob and the same readout depth. There is no intrinsic “pixels are harder to reason over” penalty for legible content.
- **It is generation under load.** A single pass cannot simultaneously perceive, reason, and write a long faithful review, so it sacrifices and drops $\sim 1/3$ of what it saw. Decomposition gives perception a dedicated externalization pass and synthesis a dedicated, *full-budget* pass over complete text. Collapsing the two into one call crashes coverage or fails to emit a review at all (the transcript exhausts the output budget), so the separation, not the modality, is the active ingredient.

We call the pattern **perceive–externalize–synthesize**. This account also predicts the two failure modes (§5): a compression bottleneck when synthesis cannot fit the transcript in its output budget (Mode A), and a training-prior leak when removing the source lets synthesis confabulate (Mode B). The latter is a genuine but secondary cost of dropping the modality, not the engine of the win.

This relates to, but is distinct from, prior single-modality work. Cuervo et al. [2] and the Cascade Equivalence Hypothesis [3] find speech-LLMs internally behave like ASR→LLM cascades, and Step-Audio-R1 [5] shows audio LLMs reasoning over a *textual surrogate* of the audio. Our activation-level tests find *no* reasoning-substrate penalty for propositional content in *either* modality (§4): a VLM reasons over an image of text as well as over text (E6), and, once the speech is intelligible, an omni model reasons over spoken facts as well as written ones (E7), with an apparent audio gap vanishing when low-quality TTS is replaced by a clean neural voice. The Step-Audio-R1 effect is best read as task-payload-specific, not a contradiction: its tasks (music, environmental sound) have *acoustic* answers that transcription discards. On documents, DocVLM [6] reports OCR-augmented VLMs beating pure-vision VLMs at fixed budget, while vision-only configurations dominate on GUI tasks [9, 10]. Few cross-modality same-weights measurements exist, and fewer still isolate *why*. This paper provides both, focused on **long-form review generation**, the deployment case where end-to-end models

are observably weak (they drop content and embellish) and two-pass decomposition is the obvious alternative.

Findings. (i) The two-pass decomposition (C_1) beats end-to-end (C_0): across 21 Gemini cells it reduces hallucinations on 18/21 (mean -3.86 , $p = 0.0015$) and improves coverage on 16/21 (mean $+0.118$, $p = 0.027$), despite the synthesis pass seeing strictly less than C_0 . (ii) The gain scales inversely with the C_0 baseline ($r = -0.45$, bootstrap 95% CI $[-0.73, -0.10]$): the more the monolithic pass satisfices, the more decomposition recovers. (iii) *The mechanism is generation-side, not perception or substrate* (§4). The model’s Pass-1 transcript covers 99.9% of probes and 99.7% of C_0 -dropped probes appear in it, so perception is fine. An open-VLM logit-lens finds no text-vs-image readout gap, so substrate is fine. But collapsing the two passes into one call crashes coverage, because the output budget cannot hold transcript + review. (iv) The same account predicts the two failure modes: a compression bottleneck when synthesis cannot fit the transcript in its budget (Mode A) and a training-prior leak when removing the source lets synthesis confabulate (Mode B), replicated across four long Claude cells. (v) A quote-grounded iterative-refinement variant (C_{liter} , verified at $n=5$) re-grounds synthesis against Mode B: it flips the Natural Vibration counter-cell to a clean Pareto win, trades coverage for faithfulness on other long cells, and is net-negative where Mode A compression dominates. (vi) Direction agreement between GPT-5.4 and Claude judges is 19/21 on coverage and 14/21 on hallucinations, with per-probe agreement median 0.94.

Contributions. (1) A 21-cell same-weights measurement of end-to-end (C_0) vs. two-pass decomposition (C_1), across two modalities, three vendors, and two judges. (2) *A seven-experiment mechanistic study* (§4) that adjudicates among perception, substrate, and generation-side accounts, ruling the first two out and supporting **perceive-externalize-synthesize**: perception coverage vs. surfacing (E3), an open-VLM cross-modal logit-lens under load (E6), an omni-model audio analogue (E7), the single-call collapse test (E2), a both-modalities control (E1), and DPI/native-text controls (E4/E5, reported with their confounds). (3) A two-mode account of where decomposition breaks down (Mode A compression, Mode B training-prior leak) with deployment triggers, derived from the mechanism. (4) An iterative-refinement variant (C_{liter}) that re-grounds synthesis with verbatim quotes. (5) Systematic robustness checks ($n=5$ multi-seed on four cells, inter-judge on all 21, probe-circularity on 6, reference-bias on 4), with all retract/preserve verdicts reported transparently.

2 Protocol

For each source (audio or scanned paper): (1) A **reference transcript** is produced by Gemini 3.1 Pro ASR/OCR with chunking (30-min audio chunks, 1-page-per-call OCR, Appendix A). (2) C_0 *reason-over-modality* (end-to-end): the model under test reads the raw modality and writes a review article in one call. (3) C_1 *reduce-to-text-then-reason* (the same-weights cascade): the model under test produces its own transcript (Pass-1, same prompt as the reference), then writes the review from that transcript (Pass-2, text-only). The only thing that changes between C_0 and C_1 is whether the reasoning step sees pixels/audio or text. Figure 1 schematizes the two configurations.

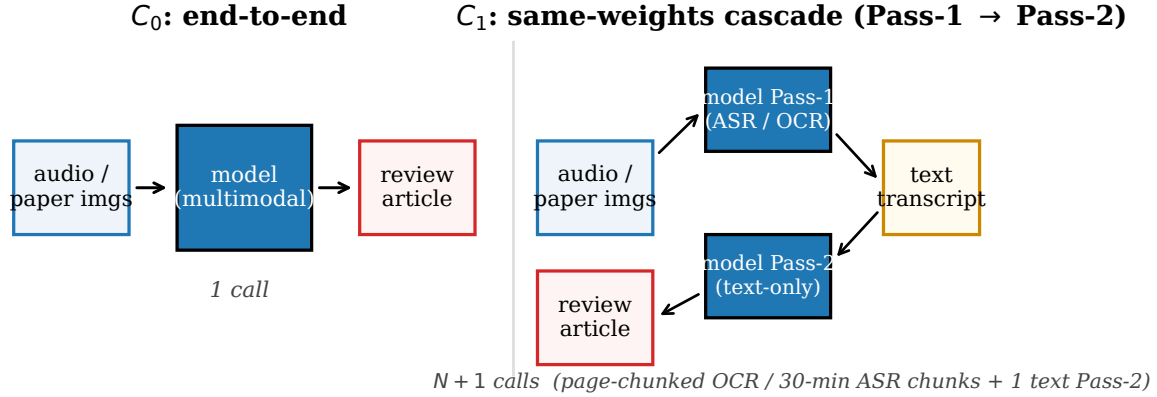


Figure 1. Reason-over-modality (C_0) vs. reduce-to-text-then-reason (C_1), same weights. C_0 is one multimodal call: the model under test reads the raw audio or page images and writes a comprehensive review, reasoning directly over the modality. C_1 first reduces the source to text (Pass-1 ASR/OCR call, same model weights), then reasons over that transcript alone (Pass-2, text-only, with the raw modality absent from the Pass-2 context). The two configurations differ only in the representation the reasoning step operates over.

Scoring. GPT-5.4 (reasoning_effort=high) acts as judge. From the reference, it extracts 30–50 atomic factual probes. For each review the judge then lists every claim not supported by the reference (*hallucination test*) and marks each probe as COVERED / MISSING (*probe coverage*). The protocol is sharper than short-form QA: the review must be both exhaustive (high coverage) and faithful (zero hallucination). Because most frontier models fail the zero-hallucination bar, we report counts and rates as operational measures.

Noise floor and statistics. We treat $|\Delta h| \leq 1$ on cells with $h < 10$, and $|\Delta \text{cov}| \leq 0.04$ (≈ 2 probes on a 50-probe cell), as within noise (presentation conventions calibrated against inter-judge direction flips, §7). For aggregate effects we report Wilson 95% CIs on coverage rates, percentile-bootstrap 95% CIs ($n_{\text{boot}}=10000$, seeded) on cross-cell means and Pearson r , and sign tests against $\Delta=0$.

Cells. 21 Gemini cells: 11 audio (Karpathy 42m, MIT 6.034 47m, Sapolsky 85m, Stanford Decarb 72m, Doudna CRISPR 48m, 3B1B Attention 26m, Veritasium Math 32m, NeurIPS Bi-nAI panel 31m, Harvard Moot Court 82m, IETF SNAC 60m, plus two meeting-minutes re-prompts of Karpathy and SNAC), and 10 papers spanning 9–176 pp: seven NASA NTRS reports (1968–1972), including Heat Pipes 104pp and SP-5100 “Shock and Vibration” 176pp, and three modern post-cutoff arXiv reviews (Fairness AI 53pp, Megagauss 75pp, Perovskite 80pp). *Cross-vendor extensions:* Claude Opus 4.7 (C_0 on Thermal Analysis 66pp, and C_1 -only on the four longest papers where Claude’s multimodal C_0 exceeds its request-size limit even with the 1M-context beta), and Mimo v2-omni (Xiaomi, via OpenRouter, on every audio source and every paper cell where the long-paper C_0 does not return empty). Reference and probes are reused from the Gemini run, isolating the question to “does the reduce-to-text advantage depend on the C_0/C_1 model?”. Full per-cell table in Appendix B.

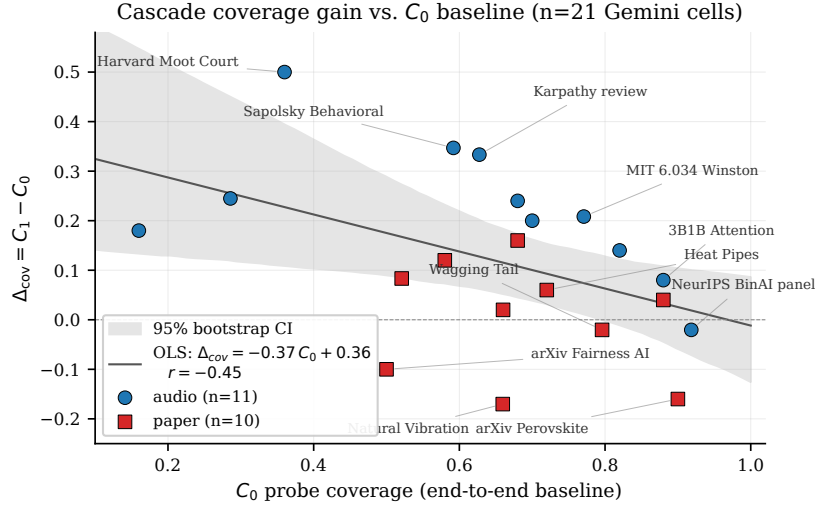


Figure 2. Reduce-to-text coverage gain vs. C_0 baseline across all 21 Gemini cells. The worse the model reasons over the raw modality (lower C_0), the more reducing to text recovers. Circles: audio. Squares: paper. OLS line with bootstrap 95% CI band, Pearson $r = -0.45$ with bootstrap CI $[-0.73, -0.10]$ (fraction $r \geq 0$: 0.9%). The five counter-cells (NeurIPS panel, Wagging Tail, arXiv Perovskite, Natural Vibration, arXiv Fairness AI) sit at high C_0 baseline (no headroom) or trigger Mode A/B (§5).

Notation. C_0 : reason-over-modality (end-to-end). C_1 : reduce-to-text-then-reason (single-pass cascade). Subscript c (C_{1c}): chunked Pass-2 (sub-summarize per chunk, then merge). Superscript concat: chunked, no merge. Superscript strip: citation surface forms replaced by neutral [REF] placeholders before Pass-2. Subscript iter: the iterative-refinement variant of §6.

3 Main result: two-pass decomposition beats end-to-end

Aggregate effect at $n = 21$. The cascade reduces hallucinations on 18/21 cells: mean $\Delta_{\text{halluc}} = -3.86$, percentile-bootstrap 95% CI $[-5.62, -2.33]$, sign-test $p = 0.0015$. It improves coverage on 16/21 cells: mean $\Delta_{\text{cov}} = +0.118$, 95% CI $[+0.048, +0.189]$, sign-test $p = 0.027$. Both bootstrap intervals exclude zero. Substituting the median across multi-judge data on the four cells with measurable judge spread (§7, Appendix G) shifts Δ_{halluc} to -3.95 $[-5.71, -2.38]$. Δ_{cov} is mechanically identical because coverage is stable under judge re-runs.

Direction of effect by baseline. Coverage gain scales inversely with the end-to-end baseline (Figure 2). This is the central prediction of the reduce-to-text account: C_0 measures how well the model already reasons over the raw modality, and the cells where it reasons worst are exactly where moving the reasoning onto text recovers the most. Excluding the two meeting-minutes re-prompts (artificially low absolute coverage because their probes use atomic-fact style while the prompt asks for structured minutes), $r = -0.47$ on $n = 19$ with CI $[-0.83, +0.10]$. The direction survives, but the CI just crosses zero on this subset.

Cascade-positive cells. The largest single coverage gain is Harvard Moot Court audio ($0.36 \rightarrow 0.86$, +50pp, multi-speaker legal argumentation with little C_0 headroom). The largest paper-cell gain is Thermal Analysis 66pp ($0.68 \rightarrow 0.84$, +16pp). Among the new

audio sources, 3B1B Attention is the cleanest hallucination win ($11h \rightarrow 0h$) and MIT 6.034 the largest combined gain ($17h/0.77 \rightarrow 4h/0.98$).

Counter-cells. The five cells where cascade fails to improve on at least one axis divide into two clusters. The first is *high-baseline / no-headroom*: NeurIPS panel $C_0=0.92$, Wagging Tail 9pp $C_0=0.80$, and arXiv Perovskite 80pp $C_0=0.90$. The second is *mode-triggered failures*: Natural Vibration 42pp ($0.66 \rightarrow 0.49$, $-17pp$, Mode B citation-prior leak) and arXiv Fairness AI 53pp ($0.50 \rightarrow 0.40$, $-10pp$, Mode A compression bottleneck). The Perovskite counter-cell shrinks from $-16pp$ at $n=1$ to $-6pp$ under $n=5$ multi-seed verification (Appendix E), so we treat it as a near-noise-floor cell going forward.

Cross-vendor: 3/3 vendors confirm cascade on a moderate paper. On Thermal Analysis 66pp (the moderate-length cell where all three vendors realize C_0 and C_1), all three improve under cascade. Gemini gives -60% halluc and $+16pp$ cov. Claude gives -64% halluc and $+8pp$ cov, reaching 1.00 probe-coverage, the dataset’s first such cell. Mimo gives $+1$ halluc (within noise) and $+24pp$ cov. Gemini and Claude converge to ≈ 1.4 hallucinations per 1k output words under cascade despite Claude producing reviews $\approx 2\times$ longer (Appendix K). End-to-end hallucination rates differ across vendors, but cascade brings them to a common rate.

Long-paper divergence. On longer cells the three vendors diverge sharply. *Heat Pipes* 104pp: Gemini cascade is small Pareto-positive ($14h/0.72 \rightarrow 11h/0.78$). Claude cascade *regresses* on faithfulness ($4h/0.90 \rightarrow 20h/0.90$), leaving C_0 on the Pareto frontier. Mimo trades $+5$ halluc for $+38pp$ coverage ($6h/0.26 \rightarrow 11h/0.64$). *SP-5100* 176pp: Gemini goes from C_0 median $46h/0.52cov$ to C_1 median $45h/0.60cov$. Claude C_1 median is $85h/0.81cov$. Claude C_0 is not realizable: 176 page-images exceed Anthropic’s per-request payload limit even with the 1M-context beta. Both Gemini and Claude C_1 sit at very high hallucination rates. Spot-checking flagged claims shows the same bibliographic-expansion signature as Heat Pipes (Appendix C). *Modern arXiv reviews (post-cutoff for Gemini 3.1 Pro)*: the cascade direction can *reverse* relative to NASA papers: Gemini Fairness AI cov $0.50 \rightarrow 0.40$, and Gemini Perovskite cov $0.90 \rightarrow 0.74$ at $n=1$ ($-6pp$ under multi-seed). We attribute the arXiv-vs-NASA inversion primarily to higher typesetting density forcing more aggressive synthesis-pass compression at the same output budget (Mode A, §5), with a possible secondary contribution from full-color rendering carrying more per-page signal. The long-paper picture motivates the mechanism study (§4) and the two regimes where decomposition stops paying off (§5).

4 Mechanism: perceive, externalize, synthesize

Why does decomposition win? Three hypotheses are consistent with the headline result but make different predictions. **H1 (substrate)**: the model reasons better over text tokens than over raw audio/image tokens, so moving the work onto text helps. **H2 (externalization / generation under load)**: a single pass cannot simultaneously perceive, reason, and write a long faithful review. Decomposition gives perception a dedicated externalization step and synthesis a dedicated, full-budget step over complete text. **H3 (attention dilution)**: raw-modality tokens are numerous and low-density, diluting the reasoning attention. We ran seven experiments (E1–E7) to separate them. The verdict is that H2 is the operative mechanism, H1 is *not* supported

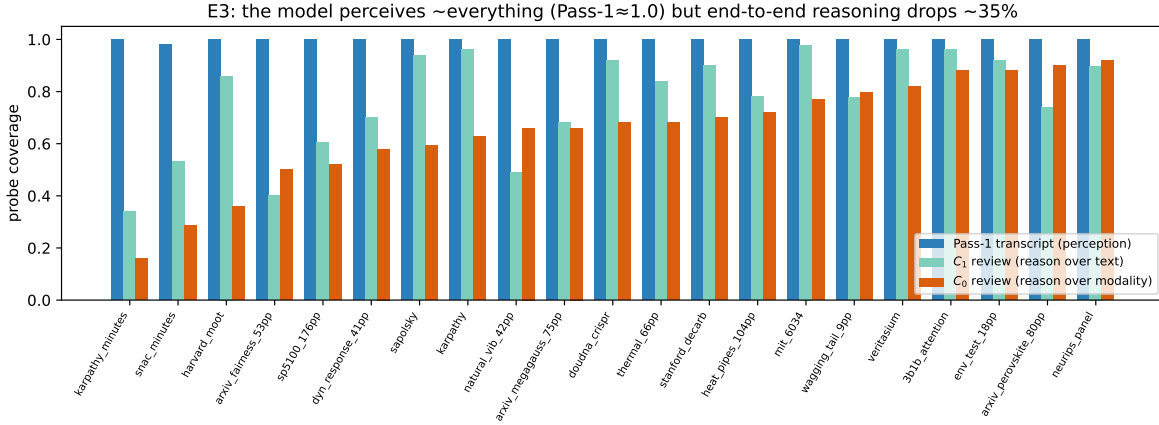


Figure 3. E3: perception is not the bottleneck. Per cell (sorted by C_0 coverage): the model’s Pass-1 transcript (blue) covers ≈ 1.0 of probes everywhere, while the end-to-end C_0 review (orange) surfaces only 0.65 on average and the C_1 synthesis (green) 0.77. Pooled, 361/362 probes that C_0 drops are present in the model’s own transcript.

(and is directly contradicted for documents), and H3 is untestable on Gemini with our tools. All experiments reuse the exact probe/hallucination judge of §2, and artifacts are released (§M). The two load-bearing generation-side contrasts (E1, E2) are run at $n=5$ seeds on all 21 cells (§4, statistical treatment). E4/E5 remain single-seed controls on an 8-cell subset and are reported as such. §4 closes with the statistical caveats, and §8 with the validation plan.

E3: the bottleneck is not perception. We scored the model’s *own* Pass-1 transcript against the same probes used for the reviews (all 21 cells). The transcript covers a mean 99.9% of probes, versus 65.2% for the C_0 review and 77.0% for the C_1 review (Figure 3). Decisively, of the 362 probes the end-to-end pass drops across the suite, 361, or 99.7% (Wilson 95% CI [98.5%, 99.9%]), are present in the model’s own transcript. The model perceives essentially everything it later omits. The content is lost *downstream* of perception. (The transcript ceiling is partly definitional, since probes are extracted from a near-identical reference transcript, but the load-bearing inference is one-directional: content the C_0 review drops is nonetheless capturable-as-text by the same weights.)

E6: text is not an intrinsically better reasoning substrate. We tested H1 directly at the activation level on an open VLM (Qwen2.5-VL-7B), controlling for prior knowledge with novel synthetic facts about nonsense entities (e.g. “the Zorblatt device weighs 7 kilograms”). Each fact is presented either as digital text or as an *image* of that same text, with a matched single-token question. A layer-wise logit-lens reads off, at every transformer layer, the log-probability and rank of the correct answer token. For an isolated fact the two modalities are indistinguishable: answer accuracy 20/20 in both conditions, final answer log-prob -0.11 (text) vs -0.05 (image), and readout depth (first layer the answer is top-1) 24.2 vs 24.5 of 28 layers. Crucially this holds *under load*: packing $K \in \{1, 4, 8, 16, 24\}$ facts into one stimulus and querying one of them, image and text remain at 1.00 accuracy with a readout-depth gap < 1 layer (TOST-equivalent within ± 1 layer) at every K (Figure 4). The same null result replicates on a second open VLM (Qwen2.5-VL-3B): 1.00 accuracy in both modalities across all

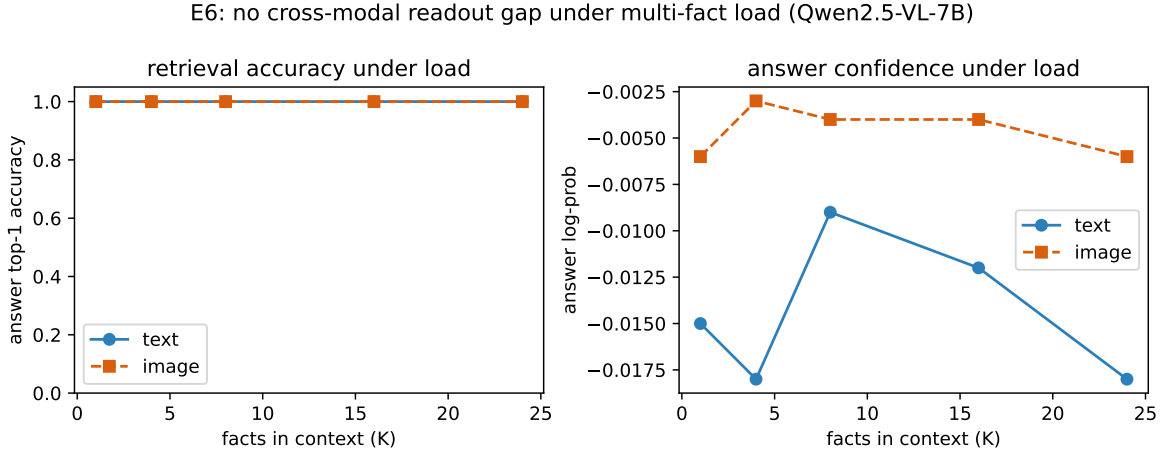


Figure 4. E6: no cross-modal substrate gap, even under load (Qwen2.5-VL-7B logit-lens). Answer accuracy (left) and answer-token log-prob (right) as a function of facts in context K , for content given as text vs as an image of that text. The curves overlap: identical propositional content is read equally well from pixels and from text up to $K=24$. The 3B model and per- K readout depths are in Appendix F.

K , depth-gap < 1 layer throughout (Appendix F reproduces the per- K ladder for both models). For legibly rendered text there is simply no cross-modal reasoning penalty: the strong form of H1 is false for documents. (Caveat: two open VLMs, 3B and 7B, on clean rendered text. The 32B variant verbalizes answers in a form our single-token matcher cannot score, so it is excluded. Frontier models and dense scanned pages remain for the validation plan.)

E7: no audio substrate gap either, once the speech is intelligible. We replicate the E6 logit-lens on an omni model (Qwen2.5-Omni Thinker), presenting each novel fact as text vs as *synthesized speech*, across three TTS engines of increasing quality (Table 1). The result is a methodological caution turned finding. With low-quality TTS (espeak, only 71% of clips intelligible to the model’s own ASR) audio *appears* to have a substrate gap (answer accuracy 0.83, readout-depth gap 90% CI [0.64, 1.78], *not* equivalent), but this tracks *intelligibility*, not modality. With two independent high-quality neural voices, *mms-tts-eng* (88% intelligible) and the more natural *fish-speech-1.5* (96%), the gap vanishes and stabilizes: audio answer accuracy 1.00 ($\geq$ text 0.96), readout depth ≈ 22.96 vs. text 22.58, and depth-gap 90% CIs [0.18, 0.57] and [0.21, 0.54], both **TOST-equivalent within ± 1 layer**. So neither documents (E6) nor audio (E7) carry an intrinsic reasoning-substrate penalty for propositional content. The cascade win is decomposition, not substrate, in both modalities. (We intended CosyVoice2 but its packaged build was unusable in our environment. The espeak-vs-clean contrast is reported because it is a cautionary result in its own right: garbled TTS manufactures a spurious substrate gap, so audio-substrate claims must control for TTS intelligibility.)

E2: the win requires a separate, full-budget synthesis pass. If decomposition merely externalizes content into one context, doing both steps in a single call should suffice. It does not. A single call instructed to first transcribe verbatim and then write the review (modality attached) *collapses*: coverage falls far below C_1 on every paper that produced a review (Thermal $0.84 \rightarrow 0.40$ and Fairness AI $0.40 \rightarrow 0.24$, Table 2), and on three cells (MIT audio, Natural Vibration, Heat Pipes) the call never emits a review at all, because the transcript consumes the

Table 1. E7 audio readout (Qwen2.5-Omni Thinker, 24 novel facts): the apparent audio substrate gap is a TTS-intelligibility artifact. “Intellig.”: fraction of clips the model’s own ASR transcribes correctly. Depth-gap = audio–text readout depth (layers). TOST SESOI ± 1 layer.

TTS	intellig.	text acc	audio acc	depth-gap 90% CI	TOST-equivalent
espeak (robotic)	0.71	0.96	0.83	[0.64, 1.78]	no (spurious gap)
mms-tts-eng (clean)	0.88	0.96	1.00	[0.18, 0.57]	yes
fish-speech-1.5 (natural)	0.96	0.96	1.00	[0.21, 0.54]	yes

entire output budget, with long documents degenerating into the TOC dot-leader loop the chunked protocol was built to avoid (§2). This collapse is confirmed at scale. At $n=5$ seeds on all 21 cells (mean 4.67 seeds/cell), single-call coverage is below C_1 by mean $\Delta_{\text{cov}} = -0.111$ (percentile-bootstrap 95% CI $[-0.170, -0.038]$), worse on 17/20 cells that emitted a review (Wilcoxon $p = 0.019$, mixed-effects single-call coefficient -0.111 , $p < 10^{-3}$), with 7 runs again emitting no review at all. The active ingredient is therefore the *decomposition into two passes with independent output budgets*, not text-in-context per se.

E1: removing the modality is a minor faithfulness bonus, not the engine. Giving the synthesis pass *both* the transcript and the raw modality (C_{1+}) does not break the benefit and sometimes maximizes coverage (3B1B C_{1+} reaches 1.00, above both C_0 and C_1 , Table 2). Consistent with E6, pixels are not harmful to content recovery. What their presence does do is re-introduce a few hallucinations relative to pure-text C_1 (e.g. 3B1B $0 \rightarrow 5h$, Perovskite $3 \rightarrow 5h$), which is the same effect Mode B names. At $n=5$ seeds on all 21 cells (mean 4.95 seeds/cell) this holds: re-attaching the modality moves coverage by only $\Delta_{\text{cov}} = +0.069$ (95% CI $[-0.001, +0.154]$, neither significantly positive nor equivalent within ± 0.04 , i.e. inconclusive) while adding hallucinations on 14/21 cells. So dropping the modality buys faithfulness at the margin, but the coverage win, the larger effect, comes from the pass structure (E2), not from removing pixels.

E4, E5: two controls that did not separate the hypotheses (reported for completeness). E4 sought to vary vision-token count via DPI on a fixed document to probe H3. But Gemini normalizes images internally, so 96 and 200 DPI yield near-identical prompt token counts (112,007 vs 111,855 on Heat Pipes). The manipulation does not move the independent variable, and H3 is untestable this way on Gemini. E5 fed native digital text (pdf to text) to a single text-only call as a clean H1 test, but on two-column arXiv PDFs pdf to text scrambles reading order, so native-text coverage is confounded downward (Fairness AI 0.34, below even C_0). Neither supports H1, so we report both with their confounds rather than omit them.

Synthesis. The evidence converges on **perceive–externalize–synthesize**. The model perceives the content (E3). Pixels-vs-text is not the reasoning bottleneck for legible documents (E6). And the recovery requires decomposing the task into two passes with independent output budgets (E2), not merely removing the modality (E1). The headline inverse correlation ($r = -0.45$) follows directly: a low C_0 baseline marks cells where the monolithic pass satisfices most, and E3 shows the dropped content is sitting in the transcript waiting to be synthesized. The two failure modes (§5) are the account’s two predicted breakdowns: synthesis-budget overflow (Mode A) and source-removal confabulation (Mode B).

Table 2. Mechanistic conditions on the 8-cell subset (hallucinations h / probe-coverage). C_0 : end-to-end (modality). C_1 : two-pass decomposition (text). C_{1+} : synthesis pass given both transcript and modality (E1). single-call: transcribe-then-review in one response (E2, “–” = no review emitted, budget exhausted). native-text: single text-only call from pdf totext (E5, arXiv only). Single-seed.

Cell	C_0 (img/aud)	C_1 (text)	C_{1+} both	single-call	native-text
MIT 6.034 (audio)	17h/0.77	4h/0.98	–	– (budget)	–
3B1B Attention (audio)	11h/0.88	0h/0.96	5h/1.00	1h/0.80	–
Wagging Tail 9pp	3h/0.80	1h/0.78	2h/0.78	5h/0.63	–
Natural Vibration 42pp	4h/0.66	5h/0.49	9h/0.79	– (budget)	–
Thermal 66pp	5h/0.68	2h/0.84	1h/0.68	3h/0.40	–
Heat Pipes 104pp	14h/0.72	11h/0.78	10h/0.66	– (budget)	–
arXiv Fairness 53pp	14h/0.50	15h/0.40	11h/0.38	11h/0.24	13h/0.34
arXiv Perovskite 80pp	2h/0.90	3h/0.74	5h/0.84	5h/0.80	4h/0.80

Statistical treatment and scope. E3 is computed over all 21 cells and 362 dropped probes (CI above). E6 is a controlled within-item comparison ($20 \text{ facts} \times 2 \text{ modalities} \times 29 \text{ layers}$, plus a load sweep over $K \in \{1, 4, 8, 16, 24\}$) on two open VLMs (Qwen2.5-VL 3B and 7B). The two generation-side contrasts, E1 and E2, are run at $n=5$ seeds on *all 21 cells* with paired per-seed deltas: the single-call collapse (E2) is large and significant (mean $\Delta_{\text{cov}} = -0.111$, bootstrap CI $[-0.170, -0.038]$, Wilcoxon $p = 0.019$, mixed-effects $p < 10^{-3}$, plus 7 no-review runs), while the smaller E1 contrast (re-attaching the modality) is directional but inconclusive (mean $\Delta_{\text{cov}} = +0.069$, CI crosses zero). E4/E5 remain *single-seed* controls on an 8-cell subset and did not separate the hypotheses. We therefore treat the mechanism as *strongly supported by E3, E6, and the multi-seed E2, corroborated by E1*, with E4/E5 inconclusive. What remains for replication is breadth, not power: frontier models, real scanned pages, and task generality beyond long-form review (§8).

5 Where decomposition breaks down: two failure modes

The perceive–externalize–synthesize account (§4) predicts exactly two ways decomposition can fail, and both appear in the data: when the synthesis pass cannot carry the externalized transcript through its output budget (Mode A), and when removing the raw source lets the synthesis pass fill gaps from prior knowledge (Mode B, the secondary cost of dropping the modality). These are the two modes below.

Mode A: compression bottleneck (3 papers). On long sources, Pass-2 has a fixed output budget ($\approx 2\text{k}$ words). When the cascade transcript is much longer than what fits, Pass-2 must compress, dropping content the same way C_0 does. Three Gemini cells exhibit this signature: Heat Pipes 104pp (Pass-1 transcript 22k words, compression ratio $\sim 11\times$, C_1 gain +6pp), arXiv Fairness AI 53pp ($\sim 15\times$, which *loses* –10pp), and arXiv Perovskite 80pp ($\sim 9\times$, –16pp at $n=1$, –6pp under multi-seed). The fix tested on Heat Pipes is **chunked Pass-2**: split the transcript into $\sim 5\text{k}$ -word chunks, summarize each, then concatenate (or merge). Heat Pipes Gemini C_{1c}^{concat} lifts coverage $0.78 \rightarrow 0.94$ (+16pp) at +5 halluc (Figure 5 left). At $n=5$ multi-seed verification (Appendix D, Table 9) the chunked-Pass-2 direction replicates on Fairness AI and Perovskite but *not* on Megagauss, so we report chunked Pass-2 as a 2/3

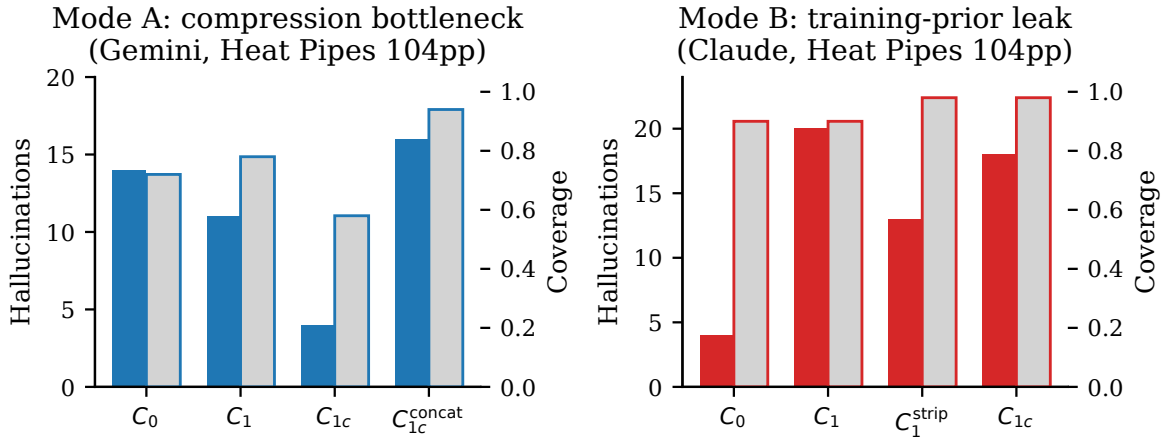


Figure 5. Two cascade failure modes on Heat Pipes 104pp. *Left:* Mode A (Gemini): Pass-2 must compress 22k→1.8k words. Chunked Pass-2 (C_{1c}^{concat}) relieves the bottleneck and lifts coverage 0.78→0.94 at +5 halluc cost (a trade along the frontier). *Right:* Mode B (Claude): C_1 hallucinates several-fold more than C_0 (4→20 on the seed plotted, $n=5$ Pass-2 mean 13, range 9–20, Appendix G). Citation-stripping (C_1^{strip}) reduces hallucinations 20→13 at $n=1$ (the $n=5$ replication retracts the magnitude, §5 and Appendix D). Halluc on left axis (lower better), coverage on right axis (higher better).

replicated Mode-A fix.

Mode B: training-prior leak (4 papers). This is the genuine cost of reduction: removing the raw source removes a grounding anchor, and on documents with heavy training-data overlap Pass-2 fills the gap from prior knowledge when the transcript abbreviates a detail. The Heat Pipes paper cites foundational references like “Reference 2 (Cotter)”. Claude’s C_1 expands these to “Cotter’s 1965 analysis (LA-3246-MS)” (Appendix L). Of the 20 unsupported claims in the analyzed Heat Pipes C_1 review (the high-end seed of the $n=5$ Pass-2 envelope, examined in Appendix L), ≈ 7 are citation-format expansions and ≈ 13 are author-name / affiliation additions (the two categories overlap on several claims), and one is a sign-error misreading. Claude C_1 hallucination counts on the four long citation-heavy papers (Pass-2 mean over available seeds, judge-median where multi-judge data exist) are: Heat Pipes 104pp ≈ 13 ($n=5$ Pass-2 mean, range 9–20), SP-5100 176pp 85 (range {47, 85, 85, 97} on the same review across four judge calls, where the original 47 was a low-end outlier), arXiv Fairness AI 23, arXiv Megagauss 17. Spot-checking the unsupported lists shows the same bibliographic-expansion signature across all four. By contrast, on the moderate citation-light Thermal 66pp paper Claude C_1 has only 4 hallucinations.

Two diagnostic signals. Two thresholds emerge as starting points for practitioner triage:

- **Mode A trigger:** Pass-1 transcript length $\gtrsim 5 \times$ Pass-2 output budget with C_1 coverage gain $< +10\text{pp}$ (Heat Pipes Gemini: $11 \times$, +6pp).
- **Mode B trigger:** input transcript contains $> \sim 2$ “Reference N ” / “Ref. N ” surface forms per 1k words *and* the topic appears in widely cited pre-cutoff literature. Heat Pipes 104pp has 55 such surface forms (2.48/1k). Thermal 66pp has 0 and shows no Mode B even on Claude.

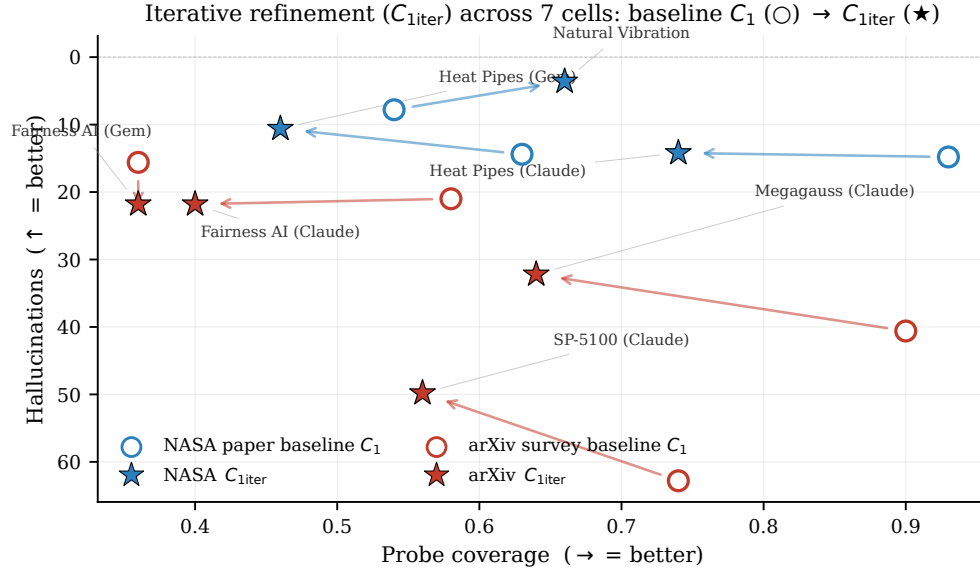


Figure 6. Iterative-refinement Pareto frontier across 7 cells ($n=5$ means). Each cell’s plain C_1 baseline (open circle) is connected by an arrow to its C_{1iter} result (filled star). Blue: NASA papers. Red: modern arXiv surveys. Only Natural Vibration moves down-and-right (fewer hallucinations *and* more coverage, a Pareto win). The other long cells move left (coverage cost from the rewrite budget), with hallucinations falling on the high-count Claude surveys (SP-5100, Megagauss) and rising on arXiv Fairness AI.

The two modes can compound (Mimo Heat Pipes shows both) or operate in isolation (Heat Pipes Gemini is dominantly Mode A, Heat Pipes Claude dominantly Mode B).

Engineering interventions tested. We evaluated two interventions across cells: *chunked Pass-2* (Mode-A target, 2/3 replication under $n=5$ multi-seed, details Appendix D) and *citation stripping* (C_1^{strip} , Mode-B target). **Citation stripping does not survive replication.** At $n=5$ Pass-2 calls on Heat Pipes Claude, C_1 hallucinations are {9, 11, 11, 15, 20} and C_1^{strip} are {5, 11, 13, 14, 14}, giving $\hat{\Delta} = -1.8 \pm 2.6$ with 95% CI $[-6.9, +3.3]$. The original “20 → 13” point estimate was within the combined Pass-2 + judge noise envelope on a vendor (Claude Opus 4.7) that deprecates temperature, so per-call determinism is not enforceable. **We retract the citation-stripping claim on Claude**, motivating the stronger structural intervention in §6.

6 Iterative refinement: a quote-grounded Mode-B fix (C_{1iter})

Pipeline. C_{1iter} is a three-step cascade variant:

1. **Pass-2a (extract).** The Pass-1 transcript is fed to the Pass-2 model with a prompt that asks for 50–150 atomic factual claims, each accompanied by a verbatim quote (≤ 25 words) from the transcript. The prompt explicitly forbids claims without a quotable span.
2. **Faithfulness filter.** Each (claim, quote) pair is string-matched against the Pass-1 transcript on the normalized form (lowercased, punctuation collapsed). Exact substring passes. Otherwise, a sliding-window `SequenceMatcher.quick_ratio` at threshold 0.85 accepts close paraphrases (handles minor OCR/ASR substitutions). Claims whose quote fails are dropped.

3. **Pass-2b (rewrite).** The surviving claims are fed back as a bullet list to a final text-only rewrite call that produces the review using *only* the listed claims, with explicit instructions to add no further facts or attributions.

Filter pass-through is high, and the discipline is in extraction. Across the 7 cells at $n=5$ (mean 83–132 extracted claims per cell), 99.8% of claims (3731/3738) pass the quote filter, with only 7 rejected. Asked for a verbatim quote, the model largely self-disciplines and stops emitting claims it cannot ground, exactly the bibliographic expansions Mode B catalogues. The filter is a tripwire that confirms extraction-step discipline rather than a post-hoc culler.

Table 3. Iterative refinement ($C_{1\text{iter}}$) on 7 cells, $n=5$ seeds at $T=0.7$ (hallucinations: mean $\pm$ sd, coverage: mean). Δ columns are vs. that cell’s plain C_1 baseline, re-measured at the same $n=5$. “Pareto”: both axes improve beyond noise. “Trade”: faithfulness gain at a coverage cost. “Net-neg”: no faithfulness gain *and* a coverage loss (or both axes worse). Hallucination counts on the high-count Claude cells (SP-5100, Megagauss) carry large GPT-5.4 judge variance (§7).

Cell (vendor)	C_1 baseline	$C_{1\text{iter}}$	Δ_{halluc}	Δ_{cov}	Verdict
Natural Vibration (Gemini)	7.8 $\pm$ 2.2 / 0.54	3.6 $\pm$ 1.4 / 0.66	−4.2	+0.13	Pareto
Heat Pipes (Gemini)	14.4 $\pm$ 2.6 / 0.63	10.6 $\pm$ 3.1 / 0.46	−3.8	−0.17	trade
Heat Pipes (Claude)	14.8 $\pm$ 3.5 / 0.93	14.2 $\pm$ 4.4 / 0.74	−0.6	−0.20	net-neg
arXiv Fairness AI (Gemini)	15.6 $\pm$ 6.6 / 0.36	21.8 $\pm$ 1.0 / 0.36	+6.2	+0.01	net-neg
arXiv Fairness AI (Claude)	21.0 $\pm$ 3.9 / 0.58	21.8 $\pm$ 3.2 / 0.40	+0.8	−0.17	net-neg
SP-5100 (Claude)	62.8 $\pm$ 23 / 0.74	49.8 $\pm$ 13 / 0.56	−13.0	−0.18	trade
arXiv Megagauss (Claude)	40.6 $\pm$ 4.6 / 0.90	32.2 $\pm$ 8.5 / 0.64	−8.4	−0.25	trade

Reading. At $n=5$ (Table 3, Figure 6) the picture is more graded than the single-seed snapshot suggested: $C_{1\text{iter}}$ reduces hallucinations on five of seven cells but lowers coverage on every cell except Natural Vibration.

- *One clean Pareto win.* Natural Vibration goes from a Mode-B counter-cell to fewer hallucinations *and* higher coverage (7.8 $\rightarrow$ 3.6h, 0.54 $\rightarrow$ 0.66), short enough that the rewrite step does not over-compress.
- *Three faithfulness-for-coverage trades.* Heat Pipes Gemini, SP-5100, and Megagauss cut hallucinations (−3.8, −13, −8.4) but lose coverage: the rewrite re-grounds claims while compressing 30–47k-word transcripts into a $\sim$ 2k-word budget.
- *Three net-negatives.* On arXiv Fairness AI (both vendors) Mode-A compression dominates and the rewrite does not reduce hallucinations (on Gemini it adds them, +6). On Heat Pipes Claude the single-seed −5h faithfulness gain shrinks to −0.6h at $n=5$, giving no net gain and only a coverage loss.

Two single-seed verdicts did not survive multi-seed: the Heat Pipes Claude Mode-B “fix” (now within noise, echoing the citation-strip retraction of §5) and the “hallucinations increase” reading on SP-5100/Megagauss. Both counts *fall* at $n=5$. The single-seed rises were high-variance GPT-5.4 judge draws on 40–90-claim lists (§7).

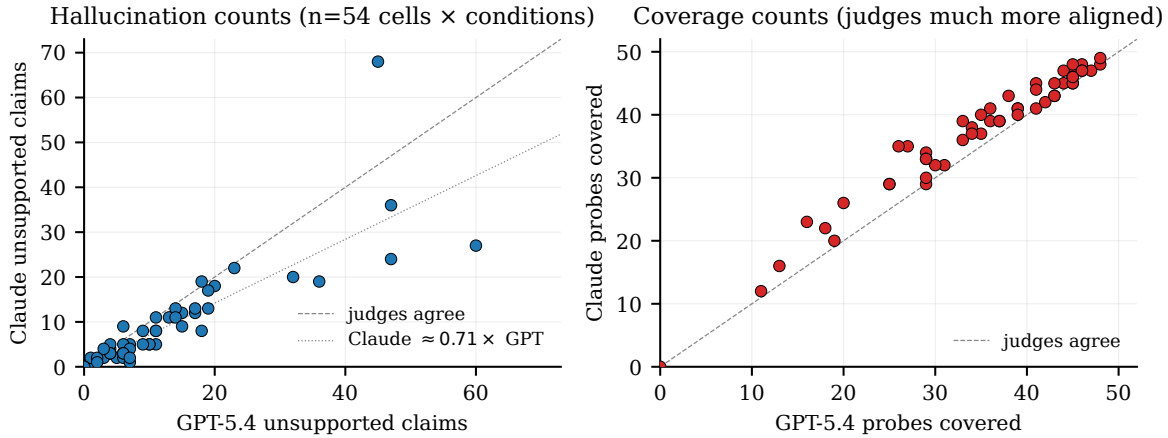


Figure 7. Inter-judge agreement across the 21 Gemini cells plus the Thermal-Claude cross-vendor cell (22 cell-condition pairs, GPT-5.4 vs Claude as judge). Dashed: identity ($y = x$). Dotted: Claude $\approx 0.71 \times$ GPT-5.4 (the median per-pair ratio). Hallucination counts (left) show Claude systematically reporting fewer unsupported claims than GPT-5.4 with most pairs along the $0.71 \times$ line. The two outliers (top-left) are the SP-5100 and Megagauss high-count cells. Probe-coverage counts (right) sit nearly on identity, consistent with per-probe agreement median 0.94.

Mechanism. C_{liter} targets Mode B, since a verbatim-quote requirement rules out unsupported expansions, but it does not relieve Mode A: the rewrite step itself compresses, so coverage falls on every long cell. The net verdict is the balance of recovered faithfulness against that compression cost. It is Pareto-positive only when the source is short enough for the rewrite to re-emit it (Natural Vibration). It is a trade when Mode-B leak is heavy enough to pay for the coverage cost (Heat Pipes Gemini, SP-5100, Megagauss). And it is net-negative when Mode-A compression dominates or the Mode-B gain washes out at $n=5$ (Fairness AI, Heat Pipes Claude). We claim it suppresses Mode B where Mode B binds, not that it is universal.

Connection to retrieval-grounded generation. The pipeline is structurally analogous to retrieval-augmented generation but operates entirely *within* the cascade: “retrieval” is the Pass-1 transcript and “attribution” is the verbatim quote. Compared to citation stripping (§5, retracted), C_{liter} targets the same Mode-B failure with a stronger structural constraint: stripping anonymizes the input citation surface form whereas C_{liter} requires the output claim to be quote-attributable. The trade-off is the rewrite-step compression cost, which is why efficacy is conditional on the cell’s dominant failure mode.

7 Robustness

We ran four independent robustness checks. Full tables are in Appendices H and I.

Inter-judge replication on all 21 Gemini cells. Claude Opus 4.7 re-scored every C_0 and C_1 review using GPT-5.4’s prompts (Figure 7). Direction agreement: **coverage 19/21 (90%), hallucination 14/21 (67%)**. Per-probe agreement: 0.82–1.00 (median 0.94). Disagreements concentrate on (i) near-tie cells where a ± 1 judge difference flips direction (NeurIPS panel, Wagging Tail) and (ii) high-count cells where the two judges agree “many claims are unsupported” but

differ in absolute count by 10+ (SP-5100, Megagauss). Claude systematically reports fewer hallucinations than GPT-5.4 (median ratio 0.71). The Mode B cross-vendor finding should therefore be read as “GPT-5.4 sees 17–85 Claude C_1 unsupported claims, while Claude judges those same Claude reviews at 13–36, still consistent with the bibliographic-expansion signature, at lower absolute counts.”

Probe-circularity check (6 cells). Probes are originally extracted by GPT-5.4, which also scores them. We re-extracted probes via Claude on six cells spanning the cascade-direction spectrum (largest audio win, near-noise paper, Mode-A and Mode-B counter-cells, high-baseline saturated, low-baseline long-paper) and re-scored the existing C_0/C_1 reviews. Under *GPT-5.4 judge*, the cascade Δ_{cov} sign is preserved on 6/6 (magnitude shifts ≤ 0.10 pp on all). Under *Claude judge* the sign flips on 1/6 (Fairness AI $-0.10 \rightarrow +0.04$, within noise). The Mode-A loss on Fairness AI and Mode-B loss on Natural Vibration both survive both substitutions.

Reference-bias check (4 cells). The Gemini reference advantages models that share its transcription priors. We re-transcribed two audio cells with Whisper-base (3B1B Attention, Karpathy) and re-OCR’d two paper cells with EasyOCR (Wagging Tail, Thermal Analysis), re-extracted probes via Claude, and re-judged with both judges. Cascade-coverage direction is preserved on 4/4 under both judges. Cascade-hallucination direction is preserved on the two audio cells (magnitudes shrink: Karpathy $-10 \rightarrow -4$ on GPT) and on Wagging Tail, but *flips* on Thermal Analysis under EasyOCR ($-3 \rightarrow +2$): EasyOCR’s transcript is $\approx 21\%$ shorter than Gemini’s and systematically loses figure captions, so claims the cascade correctly reproduces *from the Gemini Pass-1 transcript* read as unsupported against the noisier EasyOCR reference. This is a reference-side artifact, not a cascade-side finding.

Multi-seed Pass-2 envelopes on four headline cells. We re-ran Gemini C_1 Pass-2 with five seeds at $T=0.7$ on MIT 6.034 (largest audio win), Sapolsky (longest audio), Heat Pipes (long paper key cell), and arXiv Perovskite (strongest Mode-A counter-cell). Figure 8 shows the four envelopes alongside published $n=1$ points. Three of four published $n=1$ baselines fall within the $n=5$ envelope on at least one axis. The exception is Perovskite, whose $n=1$ coverage 0.74 is below all five $T=0.7$ seeds (range $[0.78, 0.90]$, mean 0.844). The cascade-cov loss on Perovskite shrinks from -16pp at $n=1$ to -6pp under multi-seed, below the per-cell noise floor. The three cells reported as wins all preserve their direction (seed-mean cov ≥ 0.90 on each). Full multi-seed tables in Appendix E.

Three sources of stochasticity. Re-runs separate three independent noise sources. (a) *Pass-2 generation variance.* Claude Opus 4.7 deprecates temperature so per-call determinism is not SDK-enforceable, and Heat Pipes Claude C_1 swings $20 \rightarrow 9$ across two runs (range 11). Gemini at $T=0.0$ is mostly but not strictly deterministic: two of three Pass-2 re-runs on Heat Pipes / SP-5100 produced byte-identical output (MD5 match), and the third differed by $\Delta h = +6$. (b) *GPT-5.4 judge variance.* On low-count cells the judge is essentially deterministic (± 1 on Heat Pipes Claude across three re-judgments of the same review). On high-count cells it is substantially stochastic: SP-5100 Claude C_1 judged four times on the same review gives $\{47, 85, 85, 97\}$ (range 50), and SP-5100 Gemini gives $\{25, 45, 59\}$ on the byte-identical review.

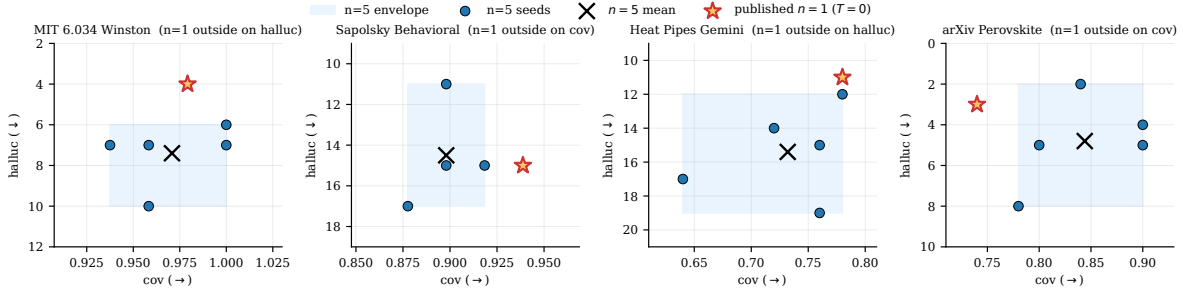


Figure 8. $n=5$ Gemini Pass-2 multi-seed envelopes for the four headline cells. Blue dots: five seeds at $T=0.7$. Black $\times$: seed mean. Orange star: published $n=1$ point at $T=0$ (red edge if outside envelope). Shaded blue rectangle: seed range. On wins (MIT 6.034, Heat Pipes Gemini) the $n=1$ value sits at the low-hallucination edge so the win direction is preserved. On Perovskite the $n=1$ is at the low-coverage edge: cascade-cov loss shrinks from -0.16 to -0.06 .

We attribute this to long unsupported-claim lists running near our 24k-token completion budget, so the judge truncates and resamples across calls. (c) *Coverage is much more stable than hallucination counts across all three sources* (HP $45 \rightarrow 45 \rightarrow 46$ probes covered, SP-5100 Claude $39 \rightarrow 39 \rightarrow 39 \rightarrow 39$). **Implications:** (i) headline aggregates use median substitution on the four high-count cells, and per-cell hallucination counts > 30 should be read with $\pm 10\text{--}25$ judge-side uncertainty. (ii) The Mode B *observation* survives all three noise sources (Claude C_1 produces many unsupported claims across every Pass-2 call and every judge re-run). The Mode B *fix* (citation stripping) does not, which motivated $C_{1\text{iter}}$ in §6. Detailed decomposition in Appendix G.

A simple held-out predictor (LOOCV, $n=21$). A leave-one-out logistic-regression predictor on four cell-level features (C_0 probe coverage, $\log_{10}$ source words, C_0 unsupported-claim count, and a paper/audio dummy) reaches LOOCV AUC = 0.65 on $\text{sign}(\Delta_{\text{cov}} > 0)$. All four coefficient signs match the qualitative picture (higher baseline reduces cascade benefit, longer source reduces benefit, more C_0 hallucinations *increases* benefit, paper modality harder than audio). AUC 0.65 is meaningful but well below deployment-ready, so we report it as an honest $n=21$ baseline. Adding a citation-density feature drops AUC to 0.61: the residual Mode-B structure is not captured by a single density feature at this sample size.

8 Discussion

Cascade cost. C_0 is one multimodal call. C_1 is $N + 1$ calls for an N -page paper (chunked OCR + a single text Pass-2) or $N_{\text{audio}} + 1$ calls for an audio source split into N_{audio} 30-min chunks. Call-count ratios in our cells range from $\approx 2\times$ (3B1B 26-min audio) to $\approx 177\times$ (SP-5100 176pp). Output-token volume scales similarly: the Pass-1 transcript is roughly as long as the source. The 21-cell evidence says cascade pays off on long-form review where end-to-end has measurable headroom, but buys nothing on already-saturated cells (NeurIPS panel $C_0=0.92$, Perovskite $C_0=0.90$). The full cost-vs-gain scatter is in Appendix J.

Practitioner checklist. Table 4 compresses the recommendations from our cells.

Table 4. Deployment checklist drawn from this paper’s cells.

Source / model class	Diagnosis	Recommended action
Audio talk or moderate paper (40–70pp) on a capable model	Cascade strongly Pareto-positive.	Default C_1 .
Moderate paper with Mode-B Gemini signature (Natural Vibration 42pp)	Mode B dominant, and the cell flips to Pareto-positive under $C_{1\text{iter}}$.	$C_{1\text{iter}}$.
Long paper, Mode A dominant (Heat Pipes 104pp Gemini)	Compression bottleneck.	C_{1c}^{concat} (chunked, no merge).
Long paper, Mode B dominant (Heat Pipes Claude)	Citation-prior leak, but the $C_{1\text{iter}}$ faithfulness gain washes out at $n=5$.	C_0 if it fits. $C_{1\text{iter}}$ buys only a coverage loss.
Long arXiv survey (Fairness AI, SP-5100, Megagauss)	Mode A + Mode B compound. At $n=5$, $C_{1\text{iter}}$ is net-negative on Fairness AI but a faithfulness/coverage trade on SP-5100/Megagauss. Chunked Pass-2 partially replicates.	Prefer C_0 when it fits. If cascade required, use C_{1c}^{concat} with multi-seed averaging.

Limitations. *Cross-vendor n .* The cleanest cross-vendor comparison is Thermal 66pp (3/3 vendors). On longer papers Claude C_0 exceeds Anthropic’s per-request payload cap, so the C_0 -vs- C_1 within-vendor comparison is incomplete on Claude. *Judge variance.* The inter-judge replication does not address single-judge intra-rater variance from re-runs with different random seeds. *Probe and reference bias.* Both are partially controlled (6 cells / 4 cells), and the cascade-hallucination direction on paper cells is partly bounded above by Gemini OCR’s recall. *$C_{1\text{iter}}$ judge variance.* $C_{1\text{iter}}$ is verified at $n=5$ (§6), but its two high-count Claude cells (SP-5100, Megagauss) inherit the ± 10 –25 GPT-5.4 judge variance, so their trade verdicts rest on means with wide per-seed spread. *Mimo audio bitrate.* Mimo audio cells ran at 16kbps mono (12kbps for Sapolsky) to fit OpenRouter’s 10MB limit, while the Gemini reference used 32kbps, so Mimo cells partly conflate audio quality with model weakness. *Task scope.* All measurements are on long-form review, so we make no claim about short-form QA, code, or agentic tasks. *Mechanism n .* E1/E2 are multi-seed ($n=5$) on all 21 cells and E6 is confirmed on two open VLMs (3B, 7B) on clean rendered text, but E4/E5 remain single-seed controls on an 8-cell subset (§4) and no E6/E7 test uses a frontier model or real scanned/degraded pages. The mechanism rests on E3 (all 21 cells), the multi-seed E2 collapse, and E6. The breadth gaps (model family, page realism, audio-under-load) await the replication below.

Validation plan. The load-bearing generation-side effects (E1/E2) are already multi-seed on all 21 cells and the E6 substrate null is confirmed on two open VLMs, so the remaining replication is about *breadth and external validity, not statistical power*. We scope it as follows. (1) *Model generality.* Extend the E6 logit-lens beyond the Qwen2.5-VL 3B/7B pair confirmed here to other open families (InternVL, Llama-Vision) and, where activations are accessible, frontier VLMs. Extend E7 across more omni models and higher-quality voices (e.g. CosyVoice2, whose packaged build was unusable in our environment) and push E7 into the multi-fact load regime, since E6 holds for documents under load but audio-under-load is still untested. (2) *Realistic-page stress.* Re-test E6 on real scanned pages (dense multi-column, equations, figures) and

degraded scans, where a substrate/dilution effect is most likely to appear, since E4’s DPI knob is inert on Gemini. (3) *Task generality*. Extend beyond long-form review to other generation-under-load tasks (long-document QA with many required facts, multi-document synthesis, meeting-minutes) and to a short-form control where the account predicts *no* decomposition benefit (the bottleneck only binds when output demand is high). (4) *Causal budget manipulation*. Hold the pipeline fixed and vary only the synthesis output budget to confirm Mode A is budget-bound, and ablate transcript completeness to map the coverage–faithfulness frontier.

9 Related work

Speech and documents. Prior cascade-vs-end-to-end results are often read as substrate claims. Our mechanism study suggests they are at least partly about decomposition and modality-specific recoverability. Cuervo et al. [2] document a text/speech understanding gap, and the Cascade Equivalence Hypothesis [3] shows speech-LLMs *internally* behave like ASR→LLM cascades, while X-Talk [4] finds explicitly cascaded speech-to-speech pipelines competitive. On documents, DocVLM [6] reports OCR-augmented VLMs beating pure-vision VLMs on DocVQA, while the OCR-Reasoning Benchmark [7] shows pure-OCR cascades cap below 50% where spatial layout carries meaning, ColPali [8] shows vision-only retrieval beats OCR, and GUI agents do better from raw pixels than from structured-text intermediates [9, 10]. The common thread is not “text is a better substrate” (our E6 finds no such gap for legible documents) but that decomposition helps when the task is generation-under-load and the content survives transcription, and hurts when the answer lives in layout/signal that transcription discards.

Step-Audio-R1: a contrast, not a corroboration. Step-Audio-R1 [5] is the sharpest statement of the boundary, and it is worth being precise, because it cuts against a naive reading of our claim. They observe that audio LLMs, when asked to deliberate, spontaneously reason over a *textual surrogate* of the audio (e.g. attributing musical sadness to “lyrics mention sadness” rather than to a minor key), and they treat this as the *failure* their modality-grounded reasoning distillation must correct. That is consistent with us, not contradictory: their tasks (music, environmental sound, paralinguistics) are precisely the case where the answer is in the acoustic signal, so reducing to text discards it. We study the complementary regime, long-form review, where the payload is what was *said*. Crucially, our activation tests find no reasoning-substrate gap for propositional content in *either* modality (E6 documents, E7 audio). The Step-Audio-R1 effect is therefore best read as *task-payload-specific*, since their tasks (music, environmental sound, paralinguistics) have answers that live in the acoustic signal, which transcription discards, rather than as a general claim that audio is a worse reasoning substrate. When the payload is propositional, as in long-form review, reducing to text loses nothing and the win is decomposition. Our Mode B (the synthesis pass confabulates once the source is removed) is the within-regime echo of their concern.

Cascade theory and faithfulness. Bengio’s Consciousness Prior [11] argues for sparse discrete bottlenecks between perception and reasoning, and our externalized text bottleneck is a deployment-time analogue. Discrete JEPa [12] makes a related architectural argument within vision. Mode B, where the model embellishes a citation surface form from training-time

knowledge, is a self-cascade analogue of the unfaithfulness problem that motivates RAG [13], RARR [14], and Attribution to Identified Sources [15]. Our C_{liter} variant adopts the in-context-attribution discipline of RARR but operates entirely within a same-weights self-cascade without external retrieval.

10 Conclusion

On long-form review, a same-weights two-pass decomposition beats end-to-end at $n=21$ (mean $\Delta_{\text{halluc}} = -3.86$ at $p=0.0015$, and mean $\Delta_{\text{cov}} = +0.118$ at $p=0.027$) even though the synthesis pass sees strictly less, and the benefit scales with how badly the monolithic pass satisfies ($r = -0.45$ between gain and baseline). The result survives median substitution on the four high-count judge-spread cells, probe re-extraction on 6 cells, reference re-generation on 4 cells, and multi-seed Pass-2 on four headline cells (with one $n=1$ counter-cell, Perovskite, retracted to near-noise-floor). Our mechanistic study locates the cause: not perception (99.7% of dropped content is in the model’s own transcript) and not a text-vs-image substrate gap (an open-VLM logit-lens finds none), but **generation under load**, where one pass cannot perceive, reason, and write a long faithful review at once, and only a decomposition with independent output budgets recovers the content (collapsing it into one call fails). The boundary is the account’s two predicted breakdowns: synthesis-budget overflow (Mode A) and source-removal confabulation (Mode B, replicated across four citation-heavy Claude cells), the latter targeted by the quote-grounded C_{liter} variant. The practical rule is **perceive, externalize, then synthesize**: decompose long-form multimodal review into a transcription pass and a full-budget synthesis pass. The win is the decomposition, not the modality. This is a deployment-time pattern with predictable failure modes, not a universal architecture.

Acknowledgements

This paper was produced using a voice-directed *whisper coding* workflow [1], in which the author specifies and reviews the work by voice while AI agents carry out the coding and experiments. AI tools including Pine Copilot and Claude Code with Claude Opus 4.8 were used during this research. We thank BSQL Networking for hosting the NVIDIA RTX PRO 6000 Blackwell Workstation Edition GPU.

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A Protocol details

Reference quality. Gemini 3.1 Pro is used for OCR/ASR with chunking. Single-call OCR on a 104-page paper produced a degenerate TOC dot-leader loop (113KB output, 56k “words” but only 264 real-word tokens, 6 page markers out of 104). Chunking (1 page per call, max 8192 output tokens) eliminates this failure mode. For audio we chunk into 30-min segments with 5-second overlap. We discovered a silent server-side truncation on one audio chunk (chunk 1 of IETF SNAC produced 1981 words instead of the expected ≈ 4368 , ending mid-sentence with “Uh, well, see, yeah, that’”). The protocol grew an automatic chunk-level word-count parity check after this incident.

Sources. Audio: Karpathy “State of GPT” (42 min, 2023), MIT 6.034 L1 Patrick Winston (47 min, 2010), Sapolsky Behavioral Genetics 1 Stanford (85 min, post-2026 reupload), Stanford Decarbonized Power Sector Zach Ming (72 min, 2025), Doudna CRISPR Basics (48 min, 2017), 3Blue1Brown “Attention in Transformers” (26 min, 2024), Veritasium “Strange Math” (32 min, 2025), NeurIPS Black-in-AI panel (31 min, 2021), Harvard Law School Ames Moot Court (82 min, 2025), IETF SNAC WG interim 2026-04-16 (60 min, ≈ 5 speakers, recorded post the most likely training-data cutoff). Papers: NASA NTRS IDs 19720009221 (Wagging Tail, 9pp), 19680000724 (Env Test, 18pp), 19690008405 (Dynamic Response, 41pp), 19690013408 (Natural Vibration, 42pp), 19700023812 (Thermal Analysis, 66pp), 19700025120 (Heat Pipes, 104pp), and SP-5100 (Shock and Vibration Technology, 176pp). The arXiv papers are arXiv:2605.09852 (Fairness AI, 53pp), arXiv:2605.13991 (Perovskite Tandem PV, 80pp), and arXiv:2605.11379 (Megagauss Physics, 75pp). NASA papers are rendered as page images at 200 DPI, and arXiv papers at 200 DPI full color.

Models and snapshots. The models are Gemini 3.1 Pro Preview (gemini-3.1-pro-preview), Claude Opus 4.7 (claude-opus-4-7), and MIMO v2-omni (xiaomi/mimo-v2-omni via OpenRouter). The judge is GPT-5.4 (reasoning_effort=high, completion budget 24k tokens), routed via OpenRouter due to scope restrictions on the local OpenAI key.

MIMO audio constraint. OpenRouter’s MIMO provider enforces a 10MB base64 audio input limit. Karpathy required re-encoding from 32kbps to 16kbps mono mp3 to fit, and Sapolsky at 85 min required 12kbps. The MIMO audio Karpathy result (C_0 4h/0.569, C_1 4h/0.588 across 51 probes, 1509 / 1268 review words) is within noise across the $C_0 \rightarrow C_1$ transition, so we treat it as a wash. *gpt-audio* (via OpenRouter) accepted audio input but refused text-only Pass-2 with HTTP 400 “This model requires that either input content or output modality contain audio”. The same-weights cascade is not realizable on this model, and C_0 alone scored 10h/0.765 on Karpathy.

Pareto terminology. We call a cascade condition *strictly Pareto-positive* relative to C_0 when it has both fewer hallucinations and higher coverage by amounts exceeding the noise floor. *Pareto-positive within noise* means strictly better on one axis and no worse than noise on the other. *Pareto-dominant* on a set means no other condition has lower hallucination at equal or higher coverage. These are presentation labels, not statistical claims.

B All 21 cells: raw counts and Wilson 95% CIs

Table 5. All 21 Gemini base cells. Halluc. is the raw unsupported-claim count (single judge call unless noted, with medians across multi-judge data given in brackets for high-count cells). Cov. is k/n (covered probes out of n) with Wilson 95% CI. Cells where C_1 *does not* strictly improve on at least one axis above the noise floor are marked with †. *cite/1k* = count of citation surface forms (numeric brackets + “Ref. N ” word forms) per 1k Pass-1 transcript words.

Cell	len	C_0 halluc	C_0 cov [95% CI]	C_1 halluc	C_1 cov [95% CI]	Δ_{cov}	cite/1k
<i>Audio (11 cells)</i>							
Karpathy review	42m	13	0.627 [0.490, 0.747]	3	0.961 [0.868, 0.989]	+0.333	0.00
Karpathy mtg-min	42m	4	0.160 [0.083, 0.285]	2	0.340 [0.224, 0.478]	+0.180	0.00
SNAC mtg-min	60m	14	0.286 [0.178, 0.424]	11	0.531 [0.394, 0.663]	+0.245	0.00
3B1B Attention	26m	11	0.880 [0.762, 0.944]	0	0.960 [0.864, 0.989]	+0.080	0.00
Veritasium Math	32m	14	0.820 [0.692, 0.903]	6	0.960 [0.864, 0.989]	+0.140	0.00
MIT 6.034 Winston	47m	17	0.771 [0.633, 0.866]	4	0.979 [0.890, 0.996]	+0.208	0.00
NeurIPS BinAI panel	31m	6	0.918 [0.803, 0.969]	4	0.898 [0.778, 0.957]	−0.020†	0.00
Doudna CRISPR	48m	6	0.680 [0.542, 0.792]	3	0.920 [0.812, 0.968]	+0.240	0.00
Stanford Decarb	72m	11	0.700 [0.562, 0.809]	6	0.900 [0.786, 0.957]	+0.200	0.00
Harvard Moot Court	82m	11	0.360 [0.241, 0.499]	4	0.860 [0.736, 0.932]	+0.500	0.00
Sapolsky Behavioral	85m	19	0.592 [0.450, 0.720]	15	0.939 [0.834, 0.980]	+0.347	0.00
<i>Paper, NASA NTRS 1968–1972</i>							
Wagging Tail	9p	3	0.796 [0.664, 0.885]	1	0.776 [0.641, 0.870]	−0.020†	1.42
Env Test	18p	5	0.880 [0.762, 0.944]	4	0.920 [0.812, 0.968]	+0.040†	0.00
Dynamic Response	41p	3	0.580 [0.442, 0.706]	2	0.700 [0.562, 0.809]	+0.120	1.19
Natural Vibration	42p	4	0.660 [0.517, 0.778]	5	0.489 [0.353, 0.628]	−0.170†	6.33
Thermal Analysis	66p	5	0.680 [0.542, 0.792]	2	0.840 [0.715, 0.917]	+0.160	0.36
Heat Pipes	104p	14	0.720 [0.583, 0.825]	11 [med 12, {11, 12, 17}]	0.780 [0.648, 0.872]	+0.060	2.80
SP-5100 Shock	176p	47 [med 46.5]	0.521 [0.383, 0.655]	45 [med 45, {25, 45, 59}]	0.604 [0.463, 0.730]	+0.083	4.00
<i>Paper, arXiv 2026 (post-cutoff)</i>							
arXiv Fairness AI	53p	14	0.500 [0.366, 0.634]	15	0.400 [0.276, 0.538]	−0.100†	0.03
arXiv Megagauss	75p	36 [{36, 36}]	0.660 [0.522, 0.776]	32 [med 30, {28, 32}]	0.680 [0.542, 0.792]	+0.020†	9.46
arXiv Perovskite	80p	2	0.900 [0.786, 0.957]	3	0.740 [0.604, 0.841]	−0.160†	0.15

C Cross-vendor details

SP-5100 176pp replication. NASA SP-5100 “Shock and Vibration Technology” (1972, 176pp, 47k pdftotext words, 4.00 citation surface forms / 1k words) was chosen as a second long citation-heavy NASA survey for Mode B replication. Gemini C_0 median 46h/0.52 cov $\rightarrow$ C_1 median 45h/0.60 cov (range {25, 45, 59} across three judge re-runs of the byte-identical review, i.e. judge variance, not Pass-2 variance). Claude C_1 median 85h/0.81 cov (range {47, 85, 85, 97} across four judge re-runs, where the originally reported 47h was a low-end outlier). MIMO C_1 60h/0.23 cov. Claude C_0 not realizable: 176 page-images exceed Anthropic’s per-request payload limit even with the 1M-context beta. Spot-checking Claude C_1 shows the same bibliographic-expansion signature as Heat Pipes.

Modern arXiv reviews. Gemini Fairness AI: 14h/0.50 $\rightarrow$ 15h/0.40 (−10pp coverage, Mode A). Claude C_1 23h/0.70, and MIMO C_1 18h/0.32. Gemini Megagauss: C_0 median 36h/0.66 $\rightarrow$ C_1 median 30h/0.68 (high baseline-halluc physics review). Claude C_1 17h/0.92, and MIMO C_1 19h/0.26. Gemini Perovskite: 2h/0.90 $\rightarrow$ 3h/0.74 at $n=1$ (−6pp under multi-seed, near-noise-floor). Claude C_1 0h/0.90, and MIMO C_1 2h/0.90. The arXiv pattern (Mode A on 2/3) is opposite-direction from the NASA pattern despite comparable page counts. We attribute this to (a) arXiv full-color 200 DPI rendering carrying more visual signal per page, and (b) higher

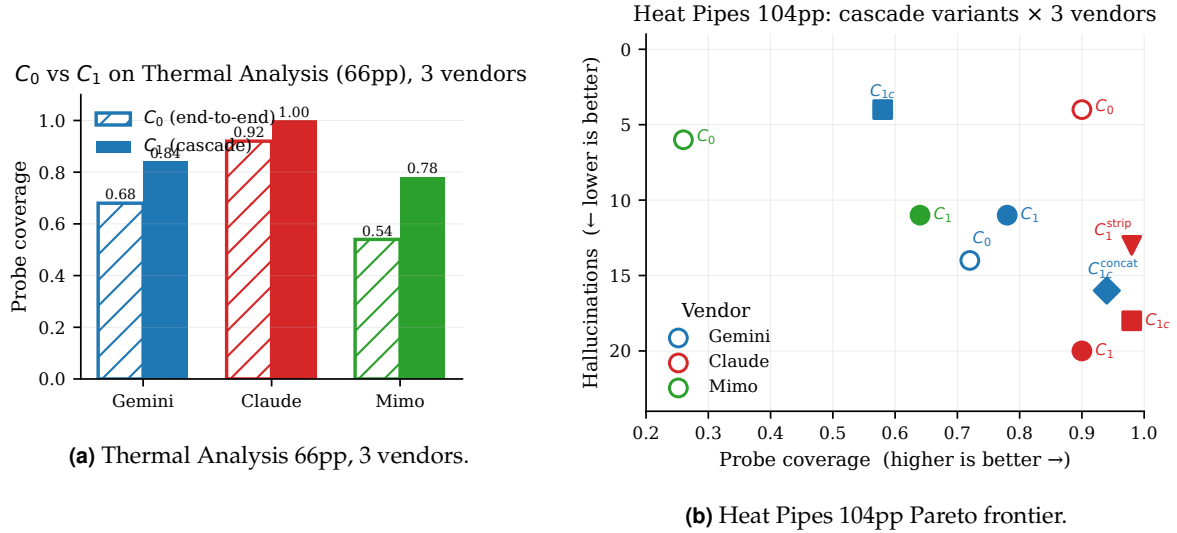


Figure 9. (a) On Thermal Analysis the cascade improves coverage on every vendor: Gemini 0.68 → 0.84, Claude 0.92 → 1.00, Mimo 0.54 → 0.78. (b) On Heat Pipes 104pp the three vendors give three patterns. Gemini cascade is small-Pareto-positive. Claude C₁ regresses on faithfulness (4h→20h) leaving C₀ on the Pareto frontier. Mimo trades +5 halluc for +38pp coverage.

typesetting density forcing more aggressive C₁ compression at the same output budget.

D Engineering interventions: chunked Pass-2 and citation stripping

Table 6. Heat Pipes Claude 104pp variants. Strip dominates plain C₁ at $n=1$ on both axes. Chunked combinations do not improve over strip alone. At $n=5$ multi-seed, the strip effect on C₁ hallucinations is statistically null ($\hat{\Delta} = -1.8 \pm 2.6$, 95% CI $[-6.9, +3.3]$).

Variant	Halluc.	Coverage	Words
C ₀ (image baseline)	4	0.900	3,308
C ₁ (single Pass-2)	20	0.900	2,077
C ₁ ^{strip} (citations → [REF])	13	0.980	2,471
C _{1c} (chunked + merge)	18	0.980	2,769
C _{1c} ^{concat} (chunked, no merge)	20	0.940	2,777
C _{1c} ^{strip} (strip + chunked + merge)	18	0.980	2,673
C _{1c} ^{strip+concat} (strip + chunked, no merge)	16	0.980	2,766

Mode-B fix replication on 3 new long Claude cells. Table 7 reports the strip intervention across four Claude cells. On SP-5100, the published single-call snapshot C₁ 47h → C₁^{strip} 24h initially looked like a clean halving, but the unstripped C₁ review judged four times yields {47, 85, 85, 97} (median 85), so the −23h effect compares one strip call against a low-end outlier of the unstripped distribution. On Megagauss, strip has essentially no effect (17 → 19h, 0.92 → 0.94 cov). On Fairness AI only 1 surface form matched our regexes, and the “stripped” Pass-2 produced a substantially different review (23 → 41h, 0.70 → 0.62 cov), revealing that Claude’s text-only Pass-2 is itself non-deterministic. **Net Mode-B-fix status: zero replicated**

cells (Heat Pipes retracted at $n=5$, Megagauss null, Fairness AI null-by-design, SP-5100 single-call-each).

Table 7. Mode B citation-strip intervention across four long Claude cells. “surface forms” = regex hits replaced with [REF].

Cell	len	surface forms	C_1 h	C_1 cov	C_1^{strip} h	C_1^{strip} cov
Heat Pipes (recall)	104p	55	20	0.900	13	0.980
SP-5100 (new)	176p	193	47	0.812	24	0.708
arXiv Megagauss (new)	75p	298	17	0.920	19	0.940
arXiv Fairness AI (new)	53p	1	23	0.700	41	0.620

Mode-A chunked Pass-2 replication and multi-seed verification. $n=1$ ($T=0$) results are consistent across four papers (Table 8): C_{1c}^{concat} (chunked, no merge) recovers coverage on long sources at a hallucination cost, while C_{1c} (chunked + merge) over-compresses: on Perovskite and Megagauss the merge variant collapses to 0.10 and 0.12 coverage respectively. At $n=5$ multi-seed (Table 9), the chunked-Pass-2 direction survives on Fairness AI and Perovskite but *not* on Megagauss, whose original $n=1$ (0.74cov) sits above the entire $n=5$ envelope $[0.64, 0.68]$ at $T=0.7$. We retract the original “chunked Pass-2 lifts coverage on all three arXiv counter-cells” claim.

Table 8. Chunked-Pass-2 ($n=1$, $T=0$) across four Gemini long-paper cells.

Cell	len	C_1 h/cov	C_{1c} h/cov	C_{1c}^{concat} h/cov	$\Delta_{\text{cov}}^{\text{concat}}$
Heat Pipes (recall)	104p	11 / 0.78	4 / 0.58	16 / 0.94	+0.16
arXiv Perovskite (new)	80p	3 / 0.74	1 / 0.10	13 / 0.82	+0.08
arXiv Megagauss (new)	75p	32 / 0.68	0 / 0.12	36 / 0.74	+0.06
arXiv Fairness AI (new)	53p	15 / 0.40	5 / 0.32	37 / 0.52	+0.12

Table 9. Multi-seed verification of C_{1c}^{concat} on the three arXiv cells ($n=5$ seeds, Gemini 3.1 Pro Pass-2, $T=0.7$).

Cell	C_1 baseline	orig $n=1$	$n=5$ halluc	$n=5$ cov	cov range	dir. survives
arXiv Fairness AI	15h / 0.40	37h / 0.52	31.6 ± 1.9	0.496 ± 0.055	$[0.440, 0.580]$	yes
arXiv Megagauss	32h / 0.68	36h / 0.74	42.2 ± 4.3	0.664 ± 0.017	$[0.640, 0.680]$	NO
arXiv Perovskite	3h / 0.74	13h / 0.82	16.6 ± 2.5	0.832 ± 0.033	$[0.800, 0.880]$	yes

Citation stripping on Gemini Natural Vibration ($n=5$ verified). At $n=1$ ($T=0$), strip moves halluc $5 \rightarrow 12$ but coverage $0.49 \rightarrow 0.81$ (+32pp). At $n=5$ ($T=0.7$, seeds 1–5): halluc $\{15, 11, 9, 8, 8\}$ with mean 10.2, and coverage $\{0.660, 0.723, 0.702, 0.511, 0.766\}$ with mean 0.672. The strip-vs- C_1 direction (cov gain at halluc cost) survives all 5 seeds. The magnitude on coverage is +18pp (mean) versus the original +32pp single-seed reading.

Table 10. Per-cell multi-seed envelopes for the four headline cells (Gemini C₁ Pass-2 at $T=0.7$, seeds 1–5). “base-in-envelope?” checks whether the $n=1$ value sits within the per-seed range.

Cell	orig $n=1$	$n=5$ halluc	$n=5$ cov	base-in-envelope?
MIT 6.034 Winston	4h / 0.98	7.4 ± 1.5	0.971 ± 0.028	cov yes; halluc NO (low)
Sapolsky Behavioral	15h / 0.94	14.5 ± 2.5	0.898 ± 0.017	halluc yes; cov NO (high)
Heat Pipes Gemini	11h / 0.78	15.4 ± 2.7	0.732 ± 0.056	cov yes; halluc NO (low)
arXiv Perovskite	3h / 0.74	4.8 ± 2.2	0.844 ± 0.056	halluc yes; cov NO (low)

E Multi-seed envelopes on the four headline cells

Two readings: (1) the published $T=0$ baselines are mostly representative of the $T=0.7$ distribution on the headline cells, since every cell’s $n=5$ envelope overlaps the published number on at least one axis. (2) Perovskite was the single seed-envelope point most far from the multi-seed mean (-0.10 pp coverage below it), and under multi-seed verification its cascade-cov loss reads -6 pp rather than -16 pp. We retain Perovskite as a near-noise-floor cell and treat its earlier “Mode-A counter-cell” label as overstated at $n=1$.

F E6 vision-model ladder and parametric load sweep

Table 11 reports the E6 logit-lens readout for both open VLMs across the load sweep $K \in \{1, 4, 8, 16, 24\}$ (facts packed into one stimulus, one queried). Answer accuracy is 1.00 for text and image at every K on both models, and the text–image readout-depth gap stays within ± 1 layer throughout (TOST-equivalent within a ± 1 -layer SESOI at every row). The Qwen2.5-VL-32B variant was excluded: it verbalizes the answer in a form our single-token matcher cannot capture (a spurious acc = 0 artifact), so its readout depths are unreliable, and the 3B and 7B models carry the claim.

Table 11. E6 cross-modal logit-lens across two open VLMs and a K -fact load sweep. *text/img depth* = first layer the answer token is top-1. *gap CI90* = 90% CI on (img–text) depth in layers. *equiv* = TOST-equivalent within ± 1 layer. Accuracy is 1.00 for both modalities in every row.

Model	K	text acc	img acc	text depth	img depth	gap CI90 (equiv)
Qwen2.5-VL-3B	1	1.00	1.00	32.00	32.00	[0.00, 0.00] (yes)
Qwen2.5-VL-3B	4	1.00	1.00	32.25	32.06	[−0.41, 0.04] (yes)
Qwen2.5-VL-3B	8	1.00	1.00	32.19	32.06	[−0.27, 0.02] (yes)
Qwen2.5-VL-3B	16	1.00	1.00	32.75	32.06	[−0.88, −0.49] (yes)
Qwen2.5-VL-3B	24	1.00	1.00	32.62	32.00	[−0.83, −0.42] (yes)
Qwen2.5-VL-7B	1	1.00	1.00	24.31	24.44	[−0.02, 0.27] (yes)
Qwen2.5-VL-7B	4	1.00	1.00	24.25	24.12	[−0.27, 0.02] (yes)
Qwen2.5-VL-7B	8	1.00	1.00	24.00	24.00	[−0.15, 0.15] (yes)
Qwen2.5-VL-7B	16	1.00	1.00	24.06	24.12	[−0.04, 0.17] (yes)
Qwen2.5-VL-7B	24	1.00	1.00	24.06	24.12	[−0.04, 0.17] (yes)

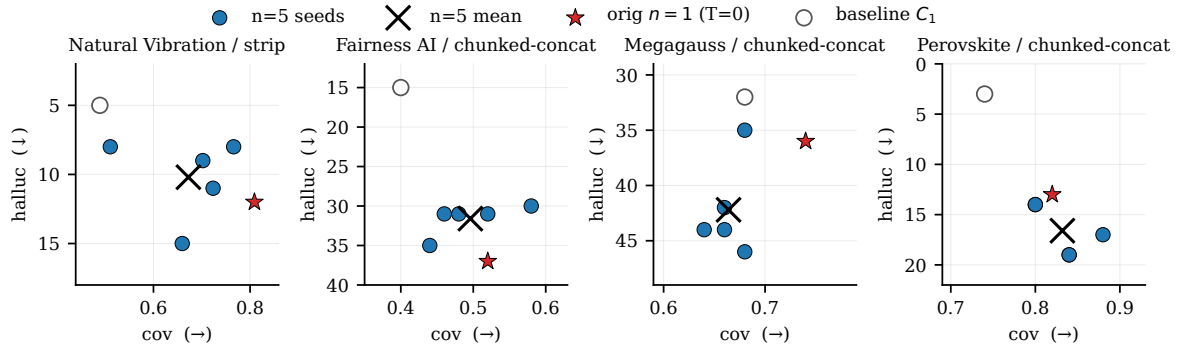


Figure 10. $n=5$ multi-seed envelopes for four intervention-verification cells. Blue dots: five Pass-2 seeds at $T=0.7$. Black $\times$: $n=5$ mean. Red star: originally-published $n=1$ at $T=0$. Open circle: cell’s C_1 baseline (no intervention). NV strip, Fairness AI, Perovskite: $n=1$ sits within $n=5$ envelope. Megagauss: $n=1$ is outside on both axes (lower halluc and higher cov than any of 5 seeds), so the published Megagauss chunked-Pass-2 cov-gain claim is retracted at $n=5$.

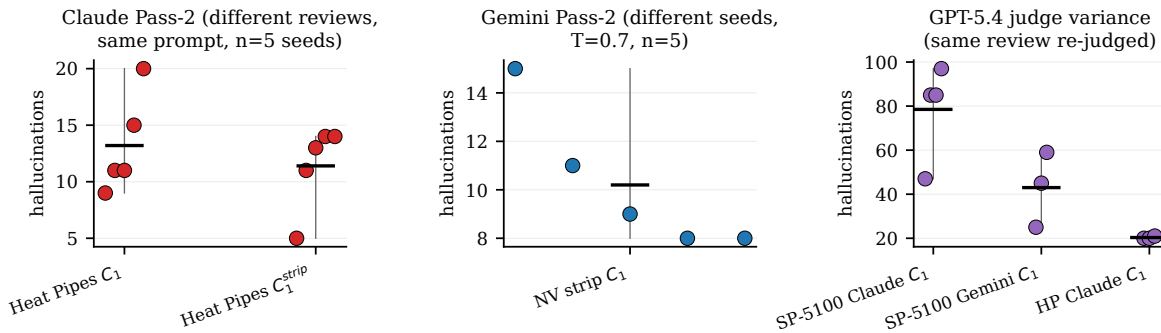


Figure 11. Three sources of stochasticity, decomposed. *Left:* Claude Pass-2 generation variance, with 5 reruns on Heat Pipes yielding $\{9, 11, 11, 15, 20\}$ (range 11). *Middle:* Gemini Pass-2 generation variance at $T=0.7$ on Natural Vibration strip: $\{15, 11, 9, 8, 8\}$ (range 7). *Right:* GPT-5.4 judge variance on the *same* review file judged multiple times. SP-5100 Claude C_1 produces $\{47, 85, 85, 97\}$ over 4 judge re-runs (range 50), while Heat Pipes Claude C_1 is stable across re-judgments. Coverage is much more stable than hallucination counts across all three noise sources.

G Three sources of stochasticity

Detailed decomposition. (a) *Claude Pass-2 generation variance.* Same input + same API + same prompt: Heat Pipes Claude C_1 swings $20 \rightarrow 9$ between two runs (strip $13 \rightarrow 14$ more stable), and the deltas are large enough that the strip effect cannot be distinguished from Pass-2 noise. Claude Opus 4.7 deprecates the temperature parameter so determinism cannot be enforced from the SDK. (b) *Gemini Pass-2 generation variance.* At temperature=0.0, two of three Gemini Pass-2 re-runs produced byte-identical output (MD5 match) on Heat Pipes / SP-5100, and the third differed in similar length (1929 vs 1815 words, $\Delta h = +6$). Gemini is therefore mostly but not strictly deterministic, consistent with GPU-level non-associativity. (c) *GPT-5.4 high-reasoning judge variance.* Essentially deterministic on low-count cells (± 1 on Heat Pipes Claude across three re-judgments) but substantially stochastic on high-count cells (± 25 on SP-5100, ± 17 on the byte-identical SP-5100 Gemini review). We attribute this to long unsupported-claim lists running up against our 24k-token completion budget, so the judge

truncates and resamples across calls. Coverage is stable on every cell across judge re-runs.

$n=5$ multi-seed verification on Heat Pipes Claude. C_1 halluc $\{9, 11, 11, 15, 20\}$ (mean 13.2, stdev 4.4, range 9–20), and C_1^{strip} halluc $\{5, 11, 13, 14, 14\}$ (mean 11.4, stdev 3.8, range 5–14). Coverage means identical (0.936 both) with similar spread. Strip effect on hallucinations at $n=5$: $\hat{\Delta} = -1.8$, $\text{SE} \approx 2.6$, 95% CI $[-6.9, +3.3]$, which crosses zero. **The Mode B fix is statistically null on Claude/Heat Pipes at $n=5$.**

Table 12. Stochasticity decomposition. Top: Claude Pass-2 generation variance (different reviews, same prompt, judged once each). Bottom: GPT-5.4 judge variance on the *same* Claude C_1 review file (one review, judged multiple times).

Cell	Condition	Run 1	Run 2	Δh	Δcov
<i>Claude Pass-2 generation variance</i>					
Heat Pipes 104p	C_1	20h / 0.90	9h / 0.90	−11	± 0.00
	C_1^{strip}	13h / 0.98	14h / 0.88	+1	−0.10
SP-5100 176p	C_1	47h / 0.81	79h / 0.81	+32	± 0.00
	C_1^{strip}	24h / 0.71	97h / 0.77	+73	+0.06
Fairness AI 53p	C_1	23h / 0.70	27h / 0.66	+4	−0.04
	C_1^{strip}	41h / 0.62	17h / 0.60	−24	−0.02
<i>GPT-5.4 judge variance (same review, judged multiple times)</i>					
Heat Pipes 104p	C_1 (calls 1–3)	20h / 0.90	21h / 0.92	+1	+0.02
SP-5100 176p	C_1 (calls 1–4)	47h / 0.81	97h / 0.81	+50	± 0.00

H Inter-judge details

Table 13. Inter-judge hallucination counts on the original 10 cells. Direction $\checkmark$ means both judges agree on the sign of $C_0 \rightarrow C_1$. (Per-cell deltas for the new 11 cells are saved in `research_log/inter_judge_summary_new12.json`.)

Cell	GPT-5.4		Claude		Direction agree?
	C_0	C_1	C_0	C_1	
Karpathy review	13	3	7	4	$\checkmark$
Karpathy mtg-min	4	2	1	0	$\checkmark$
SNAC mtg-min	14	11	10	11	$\times$
Wagging Tail (9pp)	3	1	2	2	$\times$
Env Test (18pp)	5	4	0	2	$\times$
Dynamic Response (41pp)	3	2	1	0	$\checkmark$
Natural Vibration (42pp)	4	5	2	4	$\checkmark$
Thermal Analysis (66pp)	5	2	2	1	$\checkmark$
Heat Pipes (104pp)	14	11	12	11	$\checkmark$
Thermal Analysis Claude	11	4	9	4	$\checkmark$

Per-probe agreement (the fraction of probes the two judges classify identically as COVERED / MISSING) ranges 0.882–1.000 across the (cell, condition) pairs, median ≈ 0.94 . Two of the

Table 14. Inter-judge probe-coverage counts (probes covered out of n) on the original 10 cells.

Cell	n	GPT-5.4		Claude		Direction agree?
		C_0	C_1	C_0	C_1	
Karpathy review	51	32	49	38	49	✓
Karpathy mtg-min	50	8	17	13	21	✓
SNAC mtg-min	49	14	26	15	28	✓
Wagging Tail (9pp)	49	39	38	40	38	✓
Env Test (18pp)	50	44	46	46	49	✓
Dynamic Response (41pp)	50	29	35	32	36	✓
Natural Vibration (42pp)	47	31	23	32	28	✓
Thermal Analysis (66pp)	50	34	42	34	45	✓
Heat Pipes (104pp)	50	36	39	39	41	✓
Thermal Analysis Claude	50	46	50	46	49	✓

three hallucination-direction disagreements on the original 10 cells (Wagging Tail, Env Test) are near-tie cells with ≤ 5 hallucinations under both judges. The third (SNAC) is driven by the judges assigning different C_0 baselines (14 vs 10) while agreeing on C_1 .

I Probe-circularity and reference-bias check tables

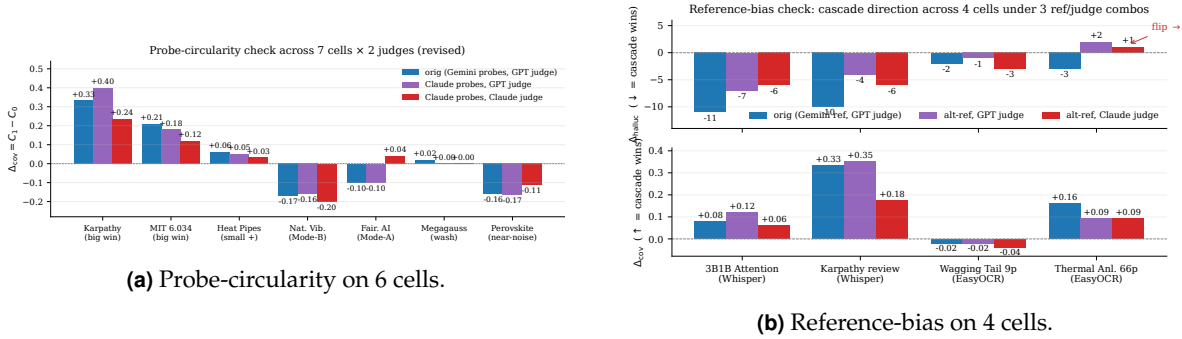


Figure 12. (a) Cascade Δ_{cov} under three (probe-extractor, judge) combinations. Sign is preserved on 6/6 under GPT-5.4 and 5/6 under Claude (Fairness AI flips, both within noise). (b) Cascade direction with non-Gemini reference (Whisper-base on 3B1B + Karpathy, EasyOCR on Wagging Tail + Thermal Analysis). Coverage direction preserved on 4/4 under both judges. Hallucination direction flips on Thermal Analysis under EasyOCR, a reference-side artifact (EasyOCR loses figure captions).

J Cascade cost

K Per-cell hallucinations per 1k output words

End-to-end hallucination rates differ per vendor (3.7 / 4.2 / 4.9 per 1k), and cascade brings Gemini and Claude to a common ≈ 1.4 per 1k floor on the moderate-length paper. Mimo does not converge to this floor.

Claude’s per-1k hallucination rate jumps from 1.2 (C_0) to 9.6 (C_1) on Heat Pipes, an $8\times$ increase that survives length normalization. The same model on the same protocol on Ther-

Table 15. Probe-circularity check on 6 cells. “Original” = published numbers (Gemini-extracted probes, GPT-5.4 judge).

Cell (orig → Claude-extracted probes)	Δ_{cov} (orig)	Claude-pr. / GPT	Claude-pr. / Claude
Karpathy review (51 → 55) [audio]	+0.333	+0.400	+0.236
MIT 6.034 (48 → 50) [audio]	+0.208	+0.180	+0.120
Heat Pipes 104pp (50 → 60) [paper]	+0.060	+0.050	+0.033
Natural Vibration 42pp (47 → 50) [paper]	−0.170	−0.160	−0.200
arXiv Fairness AI 53pp (50 → 50) [paper]	−0.100	−0.100	+0.040
arXiv Megagauss 75pp (50 → 60) [paper]	+0.020	+0.000	+0.000
arXiv Perovskite 80pp (50 → 54) [paper]	−0.160	−0.167	−0.111

Table 16. Reference-bias check on 4 cells. The non-Gemini reference rows replace OCR/ASR only, and probes are re-extracted by Claude. “orig” = Gemini reference + GPT-5.4 probes + GPT-5.4 judge.

Cell	Reference	Δ_h orig	Δ_h alt	Δ_{cov} orig	Δ_{cov} alt
3B1B Attention 26m	Whisper-base / GPT-5.4	−11	−7	+0.080	+0.120
3B1B Attention 26m	Whisper-base / Claude	−11	−6	+0.080	+0.060
Karpathy review 42m	Whisper-base / GPT-5.4	−10	−4	+0.333	+0.353
Karpathy review 42m	Whisper-base / Claude	−10	−6	+0.333	+0.176
Wagging Tail 9p	EasyOCR / GPT-5.4	−2	−1	−0.020	−0.019
Wagging Tail 9p	EasyOCR / Claude	−2	−3	−0.020	−0.038
Thermal Analysis 66p	EasyOCR / GPT-5.4	−3	+2	+0.160	+0.094
Thermal Analysis 66p	EasyOCR / Claude	−3	+1	+0.160	+0.094

Table 17. Hallucination density (per 1k output words) on Thermal Analysis 66pp.

	C_0	C_1
Gemini 3.1 Pro	4.2	1.4
Claude Opus 4.7	3.7	1.4
Mimo v2-omni	4.9	5.4

Table 18. Hallucination density (per 1k output words) on Heat Pipes 104pp. Claude C_1 at 9.6/1k is the largest value in our dataset and dwarfs Claude’s Thermal C_1 rate of 1.4/1k, supporting the Mode B interpretation as a length- and citation-density-triggered effect rather than a per-token verbosity effect.

	C_0	C_1
Gemini 3.1 Pro	7.6	6.1
Claude Opus 4.7	1.2	9.6
Claude Opus 4.7 (C_1^{strip})	—	5.3
Mimo v2-omni	7.0	6.9

mal 66pp moved 3.7→1.4. Citation stripping pulls Heat Pipes Claude back to 5.3/1k. Gemini and Mimo on the same paper sit at 6–8/1k regardless of cascade variant, indicating that the 1.4/1k floor reached on Thermal is not a property of the cascade in general, but of the specific (model, source) interaction.

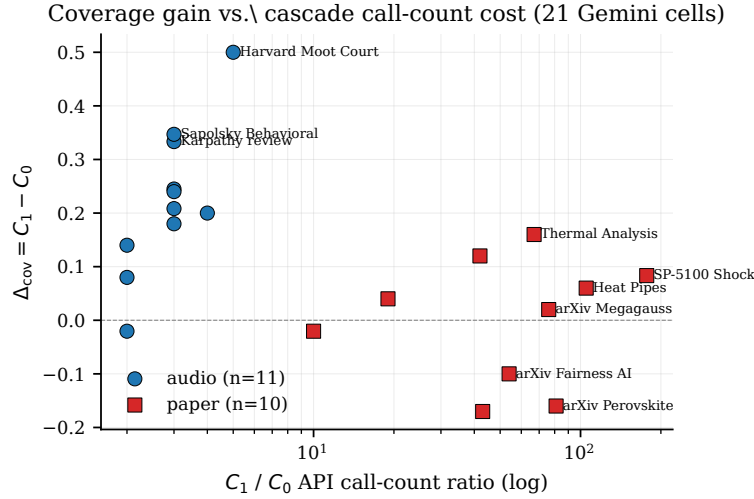


Figure 13. Cascade coverage gain vs. API call-count cost. Each point is one Gemini cell. The C_1 / C_0 call ratio for paper cells is $N + 1$ (page-chunked OCR + one Pass-2) and ranges from $10\times$ for a 9-page paper to $177\times$ for SP-5100. Audio cells use 30-min chunked ASR (ratio $\approx 2\text{--}3\times$). Cascade buys most of its coverage gain at the audio end and small or negative gains at the high-cost paper end. For example, Perovskite at $81\times$ call ratio loses 16 pp of coverage (revised to -6pp under multi-seed). SP-5100 at $177\times$ has C_0 baseline so low (0.52) that the cascade still wins on coverage ($+8\text{pp}$) but at the highest cost in the dataset.

L Specific hallucination examples (Heat Pipes Claude)

The 20 unsupported claims from Claude C_1 on Heat Pipes 104pp are dominated by training-prior expansions of bibliographic abbreviations:

- “Cotter’s 1965 analysis (LA-3246-MS)”: the transcript cites only “Reference 2 (Cotter)”, so the year and report number are training-data details.
- “correlation from Allingham and McEntire (1961)”: the transcript only says “Ref. 12”, so the author names and year are training-data details.
- “McAdams’ laminar horizontal-tube formula”: the transcript says “Reference 13 for laminar condensation on horizontal tubes”, so “McAdams” is a training-data detail.
- “Vapor chamber-fin radiator (Haller, Lieblein, and Lindow at NASA Lewis)”: the transcript says only “proposed by Haller, et al of NASA”.

One outright misreading: the transcript states the no-boiling condition as $p_v - p_\ell \leq 2\sigma \cos \theta / r_c$, but Claude’s review writes $\geq$. This is a sign error in Pass-2 reading of Pass-1 text, not training-prior leak.

M Code and data availability

All code, prompts, and per-cell judgments are released in the project repository. The pilot pipelines (pilot/pilot_audio_review.py, pilot/pilot_paper_review.py, pilot/pilot_paper_review_claude.py, pilot/pilot_paper_review_mimo.py) and post-processing scripts (pilot/chunked_pass2.py, pilot/citation_strip.py,

`pilot/iterative_refinement.py`, `pilot/inter_judge_claude.py`) reproduce every cell in the paper.